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An Asset Pricing Approach to Liquidity Effects in Corporate Bond Markets

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We use an asset pricing approach to compare the effects of the liquidity level and liquidity risk on expected U.S. corporate bond returns. Using signed transaction data, we estimate effective transaction costs for bond portfolios by a repeat-sales method. We find that the liquidity level and exposure to equity market liquidity risk affect expected bond returns. In contrast, exposure to corporate bond liquidity shocks carries an economically negligible risk premium. A simulation study shows that it is unlikely that our results are driven by measurement error in betas or multicollinearity. We present a simple theoretical model that explains these findings. (*JEL* C51, G12, G13)

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How does illiquidity affect asset prices? Theoretical models, such as the Liquidity Capital Asset Pricing Model (LCAPM) of Acharya and Pedersen (2005), suggest two potential channels. The first channel is a direct compensation for the trading costs of a given asset. In this case, illiquidity is an asset-specific characteristic leading to a liquidity level premium. The second channel focuses on exposure of asset returns to systematic liquidity shocks, leading to a liquidity risk premium. In this paper, we analyze which of these two channels is most important in the U.S. corporate bond market.

The existing literature on liquidity effects for corporate bonds has focused on one channel at a time. A first series of studies regresses credit spreads

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on bond-specific liquidity measures, thus using the liquidity level as a bond characteristic.¹ They all find evidence for a liquidity level premium: less liquid bonds have higher credit spreads. A second stream in the literature focuses on a liquidity risk premium, by investigating whether exposure of corporate bond returns to market-wide liquidity shocks in the corporate bond market or equity market is priced.² This literature generally finds evidence that liquidity risk is priced in the cross-section of corporate bonds.

The main contribution of this paper is that we integrate these two approaches in the literature by including both the effects of liquidity as a bond characteristic and various forms of liquidity risk in an asset pricing model. Given that the liquidity level and liquidity risk exposures are typically correlated, jointly analyzing their effects is necessary to get conclusive evidence. Acharya and Pedersen (2005) illustrate this for the equity market.

Determining which liquidity channel dominates is clearly important for the valuation of illiquid assets. It also provides guidance to the theoretical literature, where most existing models focus on the price effects of the liquidity level and not on liquidity risk (see, for example, Vayanos 2004; Vayanos and Wang 2012). Our results show that both the liquidity level and exposure to equity market liquidity risk have a strong effect on corporate bond prices, while the effect of corporate bond liquidity risk is insignificant and economically negligible.

Our paper has three additional contributions. First, we propose a repeat sales model, augmented for bid-ask spreads, to estimate effective transaction costs of corporate bond portfolios. Our approach is related to Edwards, Harris, and Piwowar (2007), who study the impact of transparency on bond liquidity. We argue that our measure improves upon existing measures because it can be implemented also for bonds that do not trade frequently. This is in contrast to other measures, such as the Roll (1984) measure (used by Bao, Pan, and Wang 2011) and the Amihud (2002) and Pastor and Stambaugh (2003) measures (used by Lin, Wang, and Wu 2011). Our method thus allows us to study a wider cross-section of bonds, which helps to better identify liquidity effects. In addition, we use observed trade directions at the transaction level, which implies that we can construct a liquidity measure under weaker assumptions than usual.

Second, we use a forward-looking measure of expected corporate bond returns by correcting the credit spread for the expected default losses.³ This is

¹ This literature includes Houweling, Mentink, and Vorst (2005), Covitz and Downing (2007), Nashikkar and Subrahmanyam (2006), Chen, Lesmond, and Wei (2007), Bao, Pan, and Wang (2011), and Friewald, Jankowitsch, and Subrahmanyam (2012).

² de Jong and Driessen (2012) and Acharya, Amihud, and Bharath (2013) use an equity market liquidity factor, while Chako (2005), Downing, Underwood, and Xing (2005), Lin, Wang, and Wu (2011), and Dick-Nielsen, Feldhütter, and Lando (2012) focus on a corporate bond liquidity risk factor.

³ Campello, Chen, and Zhang (2008), de Jong and Driessen (2012), and Bongaerts, de Jong, and Driessen (2011) also construct forward-looking estimates of expected corporate bond or credit default swap returns. A parallel literature argues that forward-looking expected return measures are useful for stock markets (Claus and Thomas 2001; Pastor, Sinha, and Swaminathan 2008; Chava and Purnanandam 2010).

in contrast to the “liquidity level” literature, which uses credit spreads as the left-hand-side variable. These studies do not include any risk factor exposures (such as the market beta), which may bias their estimate of the liquidity level premium. In addition, by using expected returns and an asset pricing model, we put structure on the model specification and allow for a direct interpretation of the coefficients in terms of risk exposures and risk premia. Most of the “liquidity risk” literature uses average returns to estimate expected returns. However, average bond returns critically depend on the number of defaults over the sample period. In addition, transaction data for corporate bonds are only available for short sample periods. Also, using average returns in the presence of microstructure noise may bias towards finding liquidity effects (see Asparouhova, Bessembinder, and Kalcheva 2010). The use of forward-looking expected returns requires an assumption on the term structure of expected returns and an estimate of the expected loss. We calculate the expected loss using default probability estimates from Moody’s-KMV and assumptions on the loss rate in case of default. We perform several robustness checks on this expected loss calculation.

Third, we develop a stylized theoretical model that provides a potential explanation for the absence of a corporate bond liquidity risk premium. In our model, investors prefer to trade and rebalance their portfolio using liquid assets such as equities, and avoid trading in the relatively illiquid corporate bonds as much as possible to save on trading costs. Our model then predicts that the liquidity risk associated with relatively liquid assets such as equities carries a risk premium. In contrast, exposure to liquidity risk of illiquid assets (corporate bonds) will not be priced in equilibrium since these shocks are less relevant for the investor as he avoids trading these illiquid assets when transaction costs are high. In line with this theoretical model, we empirically document that corporate bond turnover goes down in illiquid times, while equity market turnover actually increases in illiquid times.

Our key findings are as follows. First, the liquidity level substantially affects expected returns even when controlling for market and liquidity factors. This liquidity level premium is both economically and statistically significant, and quantitatively close to values predicted by theory (Amihud and Mendelson 1986). Second, we find that exposure to equity market risk and equity market liquidity risk is also priced. Third, the corporate bond liquidity risk premium is economically negligible and has the “wrong” sign in most cases. The finding that corporate bond liquidity risk is not priced is surprising, especially given existing work, but we find that our main findings survive a wide range of robustness checks.

We also perform a simulation study calibrated to our empirical sample to address two important concerns. First, both liquidity levels and betas are estimated, and the resulting measurement error may thus bias our estimates. Second, given the correlation between the liquidity level and liquidity risk beta (and other betas), multicollinearity can be a concern. Our simulation study

makes conservative assumptions about the degree of measurement error in liquidity betas and the degree of cross-sectional correlation of liquidity levels and betas. The simulations results show that even with substantial levels of measurement error and cross-sectional correlations we are able to separately identify the liquidity level and risk effects. In particular, we find that with our sample size and cross-section of triple-sorted portfolios we have sufficient statistical power to detect a corporate bond liquidity risk premium, if it were to exist.

How do our findings compare to the existing literature? Our robust finding that the liquidity level affects expected returns is consistent with the “liquidity level” literature. Four recent articles study the effects of corporate bond liquidity risk. Chako (2005), Downing, Underwood, and Xing (2005), and Lin, Wang, and Wu (2011) construct various corporate bond liquidity risk measures and show these are priced in a cross-section of corporate bond returns. There are two important differences between these three studies and our work. First, we include both the liquidity level and liquidity risk. Given that the liquidity level and liquidity risk exposure are correlated, omitting one of the two may affect the results.⁴ Indeed, if we only include corporate bond liquidity risk exposure in our regressions (thus excluding all other variables), we do find a significant corporate bond liquidity risk premium. Second, we use forward-looking expected return estimates, while the other articles use average realized returns to estimate expected returns. Dick-Nielsen, Feldhütter, and Lando (2012) regress credit spreads on liquidity variables, but also include liquidity betas as determinants of credit spread levels. However, their focus is on explaining the panel of individual credit spreads within each rating category, while our focus is to explain expected returns of portfolios sorted on credit and liquidity proxies.⁵

We also shed light on the “credit spread puzzle”. This puzzle states that credit spreads on corporate bonds are much higher than what can be justified by expected losses and exposure to market risk factors (Elton et al. 2001; Huang and Huang 2012). Previous work on this puzzle includes Cremers, Driessen, and Maenhout (2008), David (2008), and Chen, Collin-Dufresne, and Goldstein (2007). None of these articles incorporates liquidity effects. We show that liquidity effects explain a substantial part of this credit spread puzzle. This complements recent work of Bao and Pan (2013), who show that corporate

⁴ Lin, Wang, and Wu (2011) provide a robustness check where they control for the liquidity level, by multiplying Amihud’s ILLIQ measure with the turnover rate of corporate bonds and subtracting this from the average bond returns. However, the ILLIQ level itself is on a different scale than the percentage transaction costs. This is why Acharya and Pedersen (2005) transform the ILLIQ to obtain a variable that has the same level as transaction costs. Hence, the results on the liquidity level in Lin, Wang, and Wu (2011) cannot be properly interpreted.

⁵ In addition, they do not estimate an asset pricing model. For example, in their paper the coefficient on liquidity betas is estimated separately for each rating category, which results in very different coefficient estimates across rating categories.

bond volatility is higher than what is predicted by standard firm-value models (e.g., Merton 1974), and more so for less liquid corporate bonds.

Our paper is also related to the broader literature investigating liquidity effects in financial markets. In line with our findings, several articles have found that equity market liquidity risk is priced outside the cross-section of U.S. equities (see Bekaert, Harvey, and Lundblad 2007 for emerging market equities; Franzoni, Nowak, and Phalippou 2012 for the private equity market; and Sadka 2010 for hedge funds). Also related is recent work of Lou and Sadka (2011) that compares the role of liquidity level and liquidity risk in the equity market during the recent financial crisis. Finally, there is a literature on liquidity effects in government bond markets. For example, Krishnamurthy (2002) studies on-the-run versus off-the-run bonds and relates their yield difference to liquidity. Goyenko and Ukhov (2009) find a strong relation between stock and government bond liquidity. Fontaine and Garcia (2012) find evidence for a liquidity risk premium in government bond markets using a liquidity factor based on bond age.

1. Empirical Methodology and Liquidity Measures

In this section we describe our data, empirical approach, and our liquidity measures. For our analysis we use data from the TRACE Enhanced database for 2002 to 2013. We use a repeat-sales approach to measure effective transaction costs and to obtain bond returns at a portfolio level. In order to create variation in liquidity levels and liquidity betas across portfolios, we construct various triple-sorted corporate bond portfolios, sorting first on credit quality, then on liquidity proxies, and finally on liquidity betas. This sorting procedure generates independent variation in liquidity levels and liquidity risk across portfolios. In the first estimation step, we estimate exposures of these portfolio returns to equity market risk, corporate bond liquidity risk, and equity market liquidity risk. We then perform cross-sectional regressions of the forward-looking expected returns on the portfolio liquidity levels, equity market betas, and the various liquidity betas.

1.1 Asset pricing model

Our asset pricing model is a standard risk factor approach that incorporates the impact of liquidity on corporate bond prices. In a first step, we regress the time series of corporate bond portfolio returns $r_{i,t}$ (in excess of maturity-matched Treasury bond returns) on a set of risk factors F_t ,

$$r_{i,t} = \gamma_i + \beta'_i F_t + \epsilon_{i,t}. \quad (1)$$

Our forward-looking estimate of the expected excess return, $\widehat{E}(r_{i,t})$, as constructed from credit spreads corrected for expected default losses (see Section 1.5), is then regressed on the betas and the transaction costs

$$\widehat{E}(r_{i,t}) = \lambda_0 + \lambda_c c_i + \lambda'_\beta \beta_i + \alpha_i, \quad (2)$$

where c_i denotes the percentage transaction cost and α_i denotes the error term of the cross-sectional regression, which can be interpreted as the pricing error of portfolio i . The theory predicts that the intercept in this regression is zero since we focus on excess returns, but empirically we include an intercept λ_0 and test whether it equals zero. The coefficients λ_β measure the market prices of factor risk, and λ_c measures the impact of transaction costs and can, under some assumptions, be interpreted as the expected trading frequency of the asset (see Amihud and Mendelson 1986). The risk factors we include in the benchmark model are (i) equity market returns, (ii) innovations in corporate bond market liquidity, and (iii) innovations in equity market liquidity.

1.2 Portfolio selection

For our analysis we use individual bond transaction data from the TRACE Enhanced database from July 2002 up to the end of December 2013.⁶ We apply several data filters (see Appendix A for details). For each bond we calculate a yield and a credit spread by comparing the bond yield with a duration-based weighted average of the yield on two Treasury bonds with bracketing duration. We also obtain the Moody's-KMV database with expected default frequencies (EDFs). Bharath and Shumway (2008) discuss the KMV approach in detail. Although the exact procedure that KMV uses is proprietary, it is known that they use a generalization of the Merton (1974) model allowing for different debt maturities and that they map the "distance-to-default" calculated from this Merton-style model to a default rate using historical default frequencies and distances-to-default. Importantly, to calibrate the Merton model both the equity value and equity volatility are used.

As is usual in the asset pricing literature, we fit the model to portfolios rather than to individual assets. The comparison of the effects of the liquidity level and liquidity risk is reminiscent of the debate on the role of factor betas versus characteristics such as size and book-to-market in the cross-section of equity returns.⁷ We thus follow Daniel, Titman, and Wei (2001), who study the pricing of characteristics and betas in the equity market. They argue that sorting on both characteristics and betas is important to generate sufficient independent variation for these two types of variables. We therefore form triple-sorted portfolios, which are sorted first on credit quality, then on a bond liquidity proxy, and finally on the liquidity beta of a bond. To increase the number of portfolios, we sort using different variables. To conduct the credit quality sorts, we use either the S&P credit rating at the end of the previous quarter and sort into three categories (AAA-A, BBB, and BB-CCC) or the annualized default probability of the bond estimated by Moody's-KMV EDFs (tercile portfolios).

⁶ A good description of the TRACE Enhanced data can be found in Feldhütter (2012).

⁷ Key studies in this debate are Daniel and Titman (1997), Davis, Fama, and French (2000), and Daniel, Titman, and Wei (2001).

For the liquidity dimension, we sort by either amount issued, bond age, or number of trades in the previous quarter. Amount issued and age have been shown to be good proxies for liquidity by Houweling, Mentink, and Vorst (2005), while typically the number of trades will be higher for more liquid securities. In this liquidity dimension, we categorize a bond as either liquid or illiquid. The cutoff point for amount issued and age is the median, whereas for the trade count it is the 70% percentile. This proportion is required to ensure that there are enough trades in the low activity portfolio. Finally, we further sort each resulting portfolio into a high-liquidity beta and low-liquidity beta portfolio. For each combination of credit risk measure (rating/EDF) and bond liquidity proxy (amount issued, age, number of trades), we therefore have 12 portfolios ($3 \times 2 \times 2$). In total we thus obtain 72 portfolios ($2 \times 3 \times 12$). These portfolios form the basis of our tests.

The final step in the sorting procedure, where we sort on the individual bond liquidity beta, requires some more explanation. The challenge here is that many bonds do not trade very frequently and estimation of individual betas for these assets is problematic. Moreover, beta estimates for individual instruments can be rather unstable and sensitive to outliers. To deal with these issues, we use a Bayesian approach in the spirit of Vasicek (1973). Our liquidity beta for each bond is calculated as a weighted average of the direct regression estimate (obtained by regressing individual bond returns on the liquidity factor) and a portfolio-based beta. This portfolio beta is obtained by performing double sorts on credit quality and liquidity, and then using the liquidity beta of the portfolio to which the bond was assigned. The liquidity beta of these portfolios is our “best guess” (or, in Bayesian terms, “prior”) of the true liquidity beta of the bond in case insufficient trading data for this bond are available. The more precisely we can estimate the individual bond liquidity beta from transaction data, the less weight we give to these portfolio betas. This is exactly what our Bayesian solution achieves. Appendix B provides details of the procedure.

1.3 Measuring corporate bond liquidity

Like Edwards, Harris, and Piwowar (2007), we use a repeat-sales approach to measure liquidity. The main reason for using such a method is that the data are irregularly spaced: some bonds trade several times a day, but many bonds trade infrequently. The TRACE Enhanced data contains for each transaction the price, the transaction size, and an indicator of whether the transaction was buyer-initiated or seller-initiated. We thus augment the repeat-sales approach to jointly estimate portfolio returns and transaction costs using the combination of transaction prices and trade directions. For our asset pricing analysis, we want to use weekly portfolio returns, hence we apply a repeat-sales approach to all corporate bond transactions in each portfolio in each week. We now describe the procedure, with further details in Appendix C.

Consider all bonds in portfolio i in a given week. We start by modeling the k^{th} transaction of corporate bond j in portfolio i at time t_k , denoted P_{j,t_k} , as

$$P_{j,t_k} = V_{j,t_k} + c_i P_{j,t_k} Q_{j,t_k}, \text{ for all bonds } j \text{ in portfolio } i, \quad (3)$$

where V_{j,t_k} is the fundamental bond price and Q_{j,t_k} is the observed trade direction, which equals +1 in case of a buyer-initiated trade and -1 in case of a seller-initiated trade. The coefficient c_i is the effective percentage bid-ask half-spread, which we assume to be same for all bonds in portfolio i and constant within each week. This model is the same as the model of Roll (1984). Roll shows that if the fundamental value follows a random walk and trade directions are i.i.d., one can estimate c_i from the autocovariance of returns. We observe the trade directions and we can estimate c_i with weaker assumptions on the time-series behavior of the fundamental value.⁸

In our analysis, we start with hourly time intervals, but not every bond trades each hour and we therefore use a repeat-sales methodology (see, for example, Case and Shiller 1987). We take differences with respect to the previous trade of bond j and use a log approximation of Equation (3) following Edwards, Harris, and Piwowar (2007). We can then write the model as follows:

$$\ln(P_{j,t_k}) - \ln(P_{j,t_{k-1}}) = \ln(V_{j,t_k}) - \ln(V_{j,t_{k-1}}) + c_i Q_{j,t_k} - c_i Q_{j,t_{k-1}}. \quad (4)$$

We model the fundamental return as the sum of hourly latent portfolio returns $R_{i,s}$ for the portfolio i to which bond j is allocated, and a mean-zero idiosyncratic component e_{j,t_k} specific to bond j ,

$$\ln(V_{j,t_k}) - \ln(V_{j,t_{k-1}}) = \sum_{s=t_{k-1}+1}^{t_k} R_{i,s} + e_{j,t_k}. \quad (5)$$

These assumptions lead to the complete model for all data in the same portfolio,

$$\ln(P_{j,t_{j,k}}) - \ln(P_{j,t_{j,k-1}}) = \sum_{s=t_{j,k-1}+1}^{t_k} R_{i,s} + c_i Q_{j,t_k} - c_i Q_{j,t_{k-1}} + e_{j,t_k}. \quad (6)$$

As commonly done in the repeat-sales literature, this model can directly be estimated using ordinary least squares (OLS) on the panel of log-price changes, where the explanatory variables are hourly time fixed effects and the net trade directions $Q_{j,t_k} - Q_{j,t_{k-1}}$, with as regression coefficients the portfolio returns $R_{i,s}$ and transaction costs c_i . In Appendix C we discuss the econometric assumptions needed to obtain consistent estimates of transaction costs and portfolio returns.

⁸ Our model is also a special case of the Glosten and Harris (1988) model, a more general model that incorporates price impact of trading. For corporate bond markets, Edwards, Harris, and Piwowar (2007) show that large transactions actually have smaller effective costs. Therefore, we do not incorporate price impact in our model.

As mentioned above, we assume the transaction costs to be constant within every week, as the asset pricing model is estimated at a weekly frequency, but returns $R_{i,s}$ are estimated at the hourly level to control as much as possible for fundamental price changes. These hourly returns $R_{i,s}$ are then aggregated to weekly portfolio returns. In the end the repeat-sales regression thus produces a time series of weekly portfolio returns and transaction costs. We create excess returns by subtracting the return on a portfolio of Treasury bonds with the same average duration as the corporate bond portfolio.

The next step is to construct liquidity innovations from the weekly series of portfolio-level transaction costs. We assume that returns have no serial correlation, and we take the residuals of an autoregressive AR(4) model as the liquidity innovations. We also estimate the market-wide transaction costs by averaging costs across portfolios. We use the weekly AR(4) innovations in these market-wide costs as the corporate bond liquidity risk factor in our asset pricing model.⁹ We also analyze the effects of exposure to equity market liquidity. We construct this measure by taking weekly AR(4) innovations in the equally weighted mean of the Amihud (2002) ILLIQ measure across all stocks in CRSP.¹⁰

1.4 Validation of the bond liquidity measure

In Figure 1 we graph the corporate bond transaction costs obtained through the repeat-sales procedure described above, averaged each week across all portfolios. Clearly, illiquidity peaks during events in the 2008 credit crisis. We compare our measure to three previously used measures of corporate bond liquidity for validation purposes.

First, Feldhütter (2012) and Dick-Nielsen, Feldhütter, and Lando (2012) propose to measure the bid-ask spread using imputed roundtrip trades (IRT). The idea of this measure is to find trades of the same bond within a given time interval. In many cases, these trades represent a dealer acting as market maker, who buys from and sells to end users and collects a bid-ask spread as fee. For each bond, we thus take the difference between the highest price and the lowest price of trades within each trading hour as an estimate of the bid-ask spread. This measure does not use the trade directions of transactions. We then obtain an aggregate measure by averaging this bid-ask spread across all bonds, and in Figure 1 we compare this IRT measure with our measure. Clearly, the two measures are strongly related, with a correlation of 83.2%. The IRT measure peaks at a somewhat higher level during the crisis because this measure may

⁹ Note that the repeat-sales approach does not require assumptions on the time-series behavior of the portfolio return or transaction costs. Hence, once one has estimated the time series of portfolio returns with the repeat-sales model, one can subsequently specify any time-series model for these returns (such as an AR model) without generating an inconsistency between the time-series model and the repeat-sales model.

¹⁰ The equally weighted mean of the ILLIQ measure is calculated after winsorizing the cross-section of ILLIQs at the 5% and 95% levels.

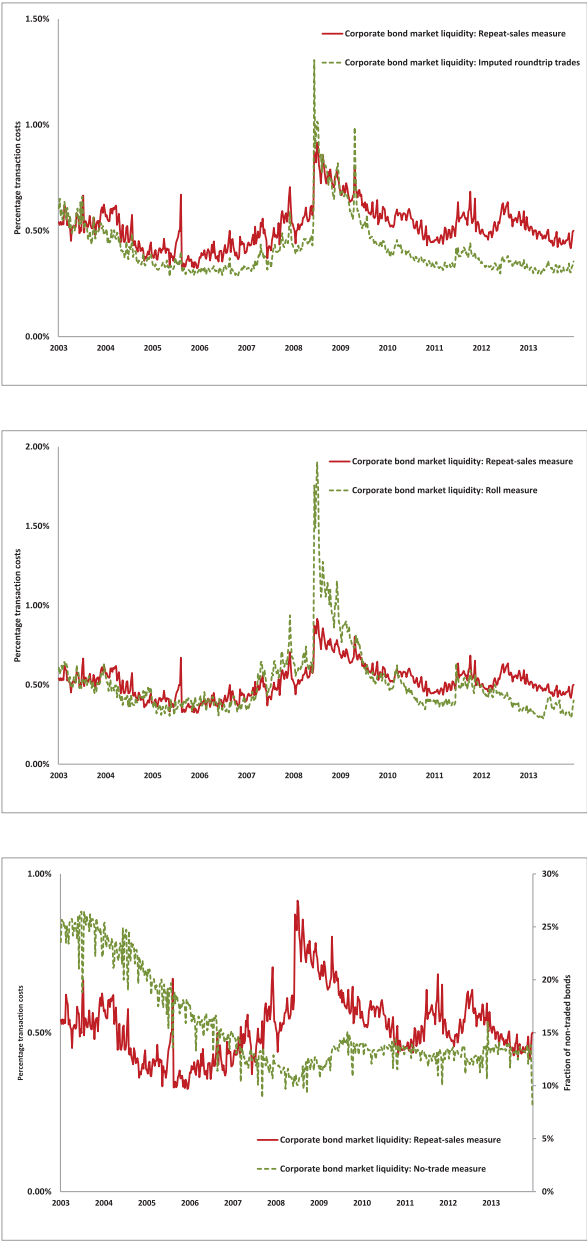


Figure 1
Time series of corporate bond transaction costs

The figure shows weekly time series of the transaction costs obtained by our repeat-sales model (solid line in each panel), averaged across all portfolios. In the upper panel, this measure is compared to the imputed roundtrip cost measure of Feldhütter (2011) and Dick-Nielsen, Feldhütter, and Lando (2011), averaged across all bonds (dashed line). In the middle panel, our repeat-sales measure is compared to the Roll measure, which is estimated following the approach of Bao, Pan, and Wang (2012). In the lower panel, the repeat-sales measure is compared to a "no-trade" measure that captures the fraction of bonds that do not trade in a given week. The sample period is 2003 to 2013.

also pick up fundamental volatility within a given trading hour. During the crisis, this fundamental volatility was clearly higher than usual.

Second, we use the standard Roll (1984) measure, where we follow as much as possible the implementation of Bao, Pan, and Wang (2011). This measure can only be constructed for frequently traded bonds and does not use the trade sign indicators. Hence it is expected to be a less precise measure of liquidity. We estimate for each week and for each bond the autocovariance of trade-by-trade returns, $Cov(\ln(P_{j,t_{k+1}}) - \ln(P_{j,t_k}), \ln(P_{j,t_k}) - \ln(P_{j,t_{k-1}}))$, and then average this weekly autocovariance across all bonds to obtain an aggregate measure. As shown by Roll (1984), if the fundamental return in Equation (4) is i.i.d., the transaction costs c equal the square root of minus the autocovariance. We thus perform this transformation for the aggregate autocovariance, and Figure 1 shows the time series of this measure along with our measure. Again, the correlation with our measure is high (80.2%). During the crisis, the Roll measure seems to overestimate transaction costs. Bao, Pan, and Wang (2011) compare the Roll measure to the estimates of Edwards, Harris, and Piwowar (2007), who use an approach related to ours, and also find that the transaction costs implied by the Roll measure are higher than more direct estimates.

Third, we use an alternative liquidity measure related to the frequency of zero returns, as used in equity markets by Lesmond, Ogden, and Trzcinka (1999) and Bekaert, Harvey, and Lundblad (2007). For each portfolio, we calculate the fraction of bonds for which we observe no transactions in a given week. Following Bekaert, Harvey, and Lundblad (2007), we use value weighting, so that our “no-trade” measure is defined as the ratio of the amount issued of all bonds that are not traded in a given week, divided by the total amount issued of all bonds in the portfolio. We obtain extremely similar results using an equally weighted measure. We average this measure across all portfolios, and Figure 1 shows the time series of this aggregate no-trade measure. Typically, between 15% to 25% of the bonds are not traded in a given week. Figure 1 also shows that the no-trade measure is slightly negatively related to our benchmark measure (correlation of -28%). The no-trade measure may thus pick up a different dimension of liquidity than the measure of effective costs.

1.5 Expected return estimates

To estimate the expected excess return $E(r_{i,t})$ on a corporate bond portfolio i , we use a forward-looking approach and take the observed credit spread and correct it for expected default losses. This procedure follows de Jong and Driessen (2012), Campello, Chen, and Zhang (2008), and Bongaerts, de Jong, and Driessen (2011).

The method works as follows. Consider bond j , and denote $\pi_{j,t}$ the cumulative default probability over the entire maturity of the bond, the loss given default L , the yield on the bond $y_{j,t}$, and the corresponding government bond yield $y_{g,t}$. We approximate the coupon-paying bond by a zero-coupon bond with maturity equal to the duration of the coupon-paying bond $T_{j,t}$.

Assuming that all default losses are incurred at maturity, the expected return of holding the bond to maturity equals $(1 + y_{j,t})^{T_{j,t}}(1 - L \cdot \pi_{j,t})$. We then annualize this number and subtract the annualized expected return on the corresponding government bond to obtain our expected excess return estimate,

$$\widehat{E}_t(r_{j,t+1}) = (1 + y_{j,t})(1 - L \cdot \pi_{j,t})^{1/T_{j,t}} - (1 + y_{g,t}). \quad (7)$$

This gives an estimate of the bond-level expected return at each point in time t . The expected excess portfolio return is then constructed each week by averaging the expected bond returns over all trades in the portfolio in that week that have an EDF available.

The loss given default L is assumed to be 60%. Default probability estimates $\pi_{j,t}$, needed to construct these expected excess returns, are constructed using the Moody's-KMV EDFs. We have data on the average one-year and five-year annualized expected default frequencies (EDFs), which capture the conditional default rate in the first and fifth years, respectively. We construct the conditional expected default frequency for every bond as the duration weighted average of the one-year and five-year EDFs. For durations longer than five years, we assume that the conditional default rate is flat beyond five years. From these bond-specific EDFs we obtain the expected cumulative default probabilities over the entire maturity of the bond ($\pi_{j,t}$). We prefer using Moody's-KMV EDFs over historical rating-based default probability estimates because we observe a strong increase in the EDFs during the crisis period of 2008–09. It is not obvious how to adjust for these new market circumstances when using historical rating-based default probability estimates.

The forward-looking measure in Equation (7) transforms the expected return of holding the bond to maturity to the next-period expected return. Hence, a key assumption is that the term structure of expected returns is flat. This is admittedly a nontrivial assumption, though related work in the equity market also assumes a flat term structure of expected returns (see Claus and Thomas 2001; Chava and Purnanandam 2010). We assess the importance of this assumption in two ways.

First, we assess the slope of the term structure of expected returns. We regress the panel of bond-level expected returns (across time and bonds) on the bond duration. We include month $\times$ portfolio fixed effects so that we focus on variation of bonds with different durations within a portfolio in each month. We find a slope coefficient of 2.5 basis points per duration-year, which we believe is economically small. Second, as discussed below, we include the portfolio duration as a control variable in our asset pricing equation and find that this has little effect on the estimated liquidity and risk premia.

2. Empirical Results

2.1 Empirical approach

To estimate the asset pricing model in Equation (2) we use a Fama-MacBeth approach. We choose this approach because it allows for time variation in

expected returns, transaction costs and betas. Since our sample period includes the 2008–09 crisis period, it is conceivable that such variation is present in the data. Our forward-looking measures of expected returns and transaction cost estimates are available on a weekly basis. However, the right-hand-side betas cannot be calculated with only one week of data. Therefore, we use weighted least squares with a two-sided kernel to estimate the beta at each point in time. The kernel function gives higher weight to observations close to the measurement date (see Ang and Kristensen 2012 for more details) and is two-sided, thus using observations before and after the measurement date. For simplicity, we use a triangular (“tent shaped”) kernel function, with linearly decreasing weights for observations further away from the measurement date, and zero weights for observations that are more than 52 weeks away.¹¹ This gives a time series of weekly betas. The second-step equation (2) is then estimated for each week using the weekly observations for the forward-looking expected returns, liquidity levels and betas. We start the Fama-MacBeth procedure in July 2003, so that we have at least one year of backward-looking data to estimate the betas (as our data start in July 2002). This way we obtain a time series of weekly liquidity and risk premium estimates, which we average over time to obtain our final estimates. Standard errors for these averages are calculated with Newey-West using 24 lags.¹²

2.2 Correlations, expected returns, transaction costs, and betas

Before presenting the asset pricing results, we present summary statistics for the key variables in the asset pricing regressions. We focus in particular on correlations between our variables in order to check for multicollinearity issues. We start by reporting the time series correlations across the three factors F_t that we include in the benchmark time series regression (1): corporate bond liquidity innovations (CBLIQ), equity market liquidity innovations (EQLIQ), and equity market (S&P 500) excess returns (EQ). Panel A of Table 1 shows that equity market liquidity and equity market returns exhibit negative correlation (−26%). Liquidity innovations across bond and equity markets have a correlation of 23%. Hence, there is some commonality in liquidity across markets, but most of the liquidity variation seems market-specific.

Next we calculate for each week in our sample the cross-sectional correlations of our benchmark explanatory variables: the liquidity level and the three multivariate beta estimates (EQ, EQLIQ, and CBLIQ). We report the time series average of these correlations in panel B. Calculating these correlations addresses potential concerns about multicollinearity in our cross-sectional regressions. Most importantly, the table shows an average correlation

¹¹ For dates close to the end date of the sample a truncated triangular kernel is used.

¹² To validate the lag length choice, we also used a Bartlett kernel and the automatic lag length selection methods proposed by Newey and West (1987), as described in the Eviews manual. These methods suggest an optimal lag length between 17 and 19 lags. The resulting standard errors of the coefficients are then slightly smaller (and the t -statistics slightly larger) than the ones reported in the tables with a Bartlett kernel with 24 lags.

Table 1
Correlation matrices

Panel A: Time series correlations of factors

	CBLIQ	EQLIQ	EQ
CBLIQ	1	0.23	−0.18
EQLIQ		1	−0.26
EQ			1

Panel B: Average of cross-sectional correlations of weekly betas and transaction costs

	c	β^{CBLIQ}	β^{EQLIQ}	β^{EQ}
c	1	−0.34	−0.45	0.53
β^{CBLIQ}		1	0.31	−0.27
β^{EQLIQ}			1	−0.59
β^{EQ}				1

Panel C: Cross-sectional correlations of full-sample betas and average transaction costs

	c	β^{CBLIQ}	β^{EQLIQ}	β^{EQ}
c	1	−0.57	−0.59	0.74
β^{CBLIQ}		1	0.50	−0.50
β^{EQLIQ}			1	−0.90
β^{EQ}				1

Panel A presents time series correlations of the three main factors in the time series regression (Equation (1)): corporate bond liquidity shocks (CBLIQ), equity market liquidity shocks (EQLIQ), and equity returns (EQ). The sample period is 2003 to 2013 with weekly observations. Panel B provides cross-sectional correlations of the multivariate factor betas and the liquidity level (transaction costs) across all portfolios. These cross-sectional correlations are calculated for each week in our sample, using the time-varying estimates for betas and costs, as discussed in Table 3 in more detail. The table presents the average of these correlations across all weeks. These correlations are relevant for the Fama-MacBeth regressions. Panel C also provides cross-sectional correlations of the betas and liquidity level, but now for the time series averages of the liquidity level and full-sample estimates of the betas. These correlations are relevant for the single cross-sectional regression of average expected returns on full-sample betas and average liquidity levels (specification “full sample” in Table 3).

of −34% between the CBLIQ beta and the liquidity level, which shows that with our triple portfolio sorts the Fama-MacBeth procedure should be able to separate the effects of the liquidity level and liquidity risk. The average correlation between the equity beta and the equity liquidity beta is −59%, which shows that separating the equity premium from the equity liquidity risk premium is somewhat harder. It is important to note that these correlation estimates are biased towards zero if there is measurement error in the beta and liquidity level estimates. Hence, the “true” degree of cross-sectional correlation might be higher than what we report in Table 1. In the simulation study in Section 2.5, we incorporate measurement error and report conservative estimates of the true correlations.

Besides the Fama-MacBeth procedure, which is our benchmark estimation approach, below we also present a robustness check where we estimate the asset pricing model using a single cross-sectional regression. In this case we regress the average expected excess returns on the betas and the average transaction costs, where these averages and the betas are estimated over the full sample period. In panel C of Table 1, we present the cross-sectional correlations of these full-sample betas and average transaction costs. These full-sample betas exhibit more cross-sectional correlation. In absolute terms

the cross-sectional correlations range from 50% (equity liquidity beta and corporate bond liquidity beta) to -90% (equity beta and equity liquidity beta). This shows that a helpful aspect for identifying the level versus risk effects is that their time series variation is different, as shown in Figure F.1 in the Online Appendix. The Fama-MacBeth procedure incorporates this feature. In other words, the repeated cross-sections used in the Fama-MacBeth procedure contain additional information compared to the single full-sample cross-sectional regression.

In Table 2 we then turn to expected returns, transaction costs, and CBLIQ betas, averaged over the full sample. We split up the seventy-two corporate bond portfolios, which are sorted on credit risk, liquidity, and liquidity beta, into four groups of eighteen portfolios, based on the liquidity proxy (high or low) and liquidity beta (high or low). The table presents averages across all portfolios within each group. The first key result in Table 2 is that the estimated one-way transaction costs are substantial, on average 0.52% across portfolios and over time. We also see that the transaction costs are higher for the low-liquidity portfolios and portfolios with higher liquidity risk (more negative liquidity beta). The second result in Table 2 is that we find large positive expected returns (in excess of government bonds), around 1.9% per year on average. We also see that the sorts on liquidity and liquidity beta generate substantial variation in expected returns, with higher expected returns for low-liquidity and high-liquidity risk portfolios.

Table 2
Summary statistics on returns, costs, and betas

Average transaction costs, returns, and betas		
Average transaction costs		
	High-liquidity ptfs	Low-liquidity ptfs
Low liquidity risk ptfs	0.46%	0.48%
High liquidity risk ptfs	0.54%	0.58%
CBLIQ beta		
	High-liquidity ptfs	Low-liquidity ptfs
Low liquidity risk ptfs	-0.013	-0.000
High liquidity risk ptfs	-0.050	-0.054
Expected return		
	High-liquidity ptfs	Low-liquidity ptfs
Low liquidity risk ptfs	1.62%	1.87%
High liquidity risk ptfs	2.07%	2.27%

The table presents statistics for the return and liquidity data, across 72 triple-sorted portfolios sorted first on rating or EDF, then on trading activity, average bond age or issue size, and finally on liquidity beta. We split up the 72 portfolios into four groups of 18 portfolios, based on the liquidity proxy (high or low) and liquidity beta (high or low). The table presents average transaction costs, liquidity betas, and annualized expected excess returns for these four groups (averaged across all portfolios within each group). The CBLIQ betas capture exposure of corporate bond returns to the CBLIQ factor. All factors are scaled to have the same standard deviation as the equity market return.

Next we turn to the betas. We scale all factors such that they have the same standard deviation as equity market excess returns, so that one can easily interpret the betas. We see that corporate bonds typically have a negative exposure to corporate bond liquidity risk: corporate bond prices decrease when there is a positive shock to market-wide transaction costs. In unreported results we also find, as expected, that the equity betas are positive, and the equity liquidity risk betas are mostly negative. Table 2 shows that the sort on liquidity betas is useful, as it indeed generates a spread in liquidity betas for portfolios sorted on this beta. The liquidity betas are quite similar for portfolios sorted on the liquidity level.

Table 2 can be used as a preview for our formal regression results below. First, we see that, when moving from the high-liquidity to low-liquidity portfolios, the transaction costs (liquidity level) and expected returns increase, while the liquidity betas do not. This, in a very informal way, suggests that the liquidity level effect is priced. Then, when moving from low-liquidity risk to high-liquidity risk portfolios, we see that liquidity betas become more negative and expected returns increase, but transaction costs increase as well. As we will see below, the increase in transaction costs turns out to be sufficient to explain this variation in expected returns, and the effect of liquidity betas is negligible.

2.3 Benchmark asset pricing results

In this subsection we present the benchmark cross-sectional asset pricing results with the betas of the three factors (CBLIQ, EQLIQ, and EQ) as determinants of expected returns, in addition to the transaction costs as portfolio characteristic using the Fama-MacBeth procedure described above.

Table 3 presents the main results. Specification (1) is our benchmark specification, which includes all three betas (CBLIQ, EQLIQ, and EQ) and the liquidity level. The first key result is that the liquidity level effect is significant and positive. The coefficient is 1.06, which implies that at the average transaction cost level of 0.52% the liquidity level premium equals about 0.55% in terms of annualized expected returns. This is a substantial effect for corporate bonds, given that their total expected return is 1.9%. In contrast, the price of corporate bond liquidity risk λ_{CBLIQ} is positive and thus has the “wrong” sign. This can be understood as follows. As expected, CBLIQ betas are mostly negative (on average we have $\bar{\beta}_{CBLIQ} = -0.03$) since bond prices decrease when illiquidity increases. One would thus require a negative price of liquidity risk ($\lambda_{CBLIQ} < 0$) to obtain a positive liquidity risk premium ($\lambda_{CBLIQ} \bar{\beta}_{CBLIQ} > 0$). The positive estimate for λ_{CBLIQ} thus implies a (counterintuitive) negative liquidity risk premium. The size of this effect is negligible; however, $\lambda_{CBLIQ} \bar{\beta}_{CBLIQ}$ is equal to -5 basis points per year.

We also find significant and intuitive effects for the equity risk premium and equity liquidity risk premium. The equity risk premium is estimated at 5.03% per year. Given an average equity beta of 0.13 across all portfolios, this contributes about 0.65% to the expected return on average. Despite its

Table 3
Risk-factor model: Fama-MacBeth regression estimates

Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	Full sample
λ_c	1.06 [3.64]	2.43 [6.61]					1.24 [2.14]	1.09 [2.83]	1.03 [2.48]	0.84 [2.24]	1.34 [2.95]
λ_{CBLIQ}	1.79 [1.76]		-11.93 [-4.97]			-3.59 [-1.75]	-6.76 [-4.70]	-1.24 [-1.04]			10.33 [7.29]
λ_{EQLIQ}	-4.46 [-2.17]			-8.83 [-3.99]						-3.42 [-1.84]	-2.70 [-0.93]
λ_{EQ}	5.03 [4.37]				8.48 [5.69]	8.87 [5.72]		6.88 [4.46]	6.40 [4.15]	5.15 [4.17]	8.44 [9.17]
Intercept	0.48 [3.88]	0.75 [3.05]	1.22 [16.40]	0.94 [8.93]	0.69 [8.98]	0.75 [10.05]	0.85 [3.13]	0.51 [3.78]	0.50 [3.39]	0.60 [3.99]	0.12 [0.52]
R^2	58.8%	27.5%	21.6%	41.5%	41.4%	46.1%	38.7%	53.6%	50.2%	56.0%	84.5%

This table presents Fama-MacBeth estimates of the risk factor model in Equation (2), from weekly cross-sectional regressions of expected excess corporate bond returns on corporate bond liquidity betas (CBLIQ), equity liquidity betas (EQLIQ), equity market returns (EQ), and transaction costs (*c*). For expected returns and transaction costs, the observed levels at each week are used for each portfolio. For weekly betas a two-sided kernel, spanning at most 52 weeks of data on both sides, is used. The table reports the time series average of the weekly estimates of the liquidity premium and risk premia. The sample period is 2003 to 2013, and the cross-section consists of 72 triple-sorted portfolios sorted first on rating or EDF, then on trading activity, average bond age or issue size, and finally on liquidity beta. Fama-MacBeth *t*-statistics are given in square brackets, calculated with Newey-West with 24 lags. The betas that go into the cross-sectional regressions are calculated with all factors scaled to have the same standard deviation as the equity market return (except for the liquidity level *c*). The final row gives the time series average of the weekly cross-sectional R^2 . The final column ("Full sample") presents estimates for a single cross-sectional regression of expected returns on transaction costs (both averaged over the full sample period) and full sample beta estimates.

correlation with equity risk, the equity liquidity price of risk is also significant with the expected negative coefficient. Multiplying with the average EQLIQ beta of -0.07 , its average effect on expected returns equals 0.31% . This finding adds to the evidence that equity market liquidity risk is priced in various markets, such as private equity (Franzoni, Nowak, and Phalippou 2012) and hedge funds (Sadka 2010).

In sum, we see that all variables have a substantial effect on expected returns, except corporate bond liquidity risk. Note that most coefficients in the cross-sectional regression have quite high t -statistics. This is a direct consequence of the use of forward-looking expected returns as left-hand-side variables, which are less noisy than average realized returns. In specifications (2) to (10), we consider various restricted versions of the benchmark specification (1). We see that the effect of the liquidity level is large and very stable across specifications. In contrast, for the corporate bond liquidity risk premium we only find an economically and statistically significant effect if we exclude the equity market risk premium. In this case, the corporate bond liquidity risk premium picks up the effect of the omitted equity risk. In Table 3 our preferred specifications include the equity risk premium, since we think any reasonable asset pricing model should at least have one market risk factor. Notice also that this equity risk premium, when included in the specification, is always statistically significant and hence models without this factor are misspecified. Finally, equity market liquidity risk is priced across all specifications.

The final specification presents results for a full-sample estimation of a single cross-sectional regression. Specifically, we run a cross-sectional regression of the average expected excess returns on the betas and the average transaction costs. These averages and the betas are all estimated over the full sample period.¹³ The results show again a strong effect of the liquidity level, while the corporate bond liquidity risk premium again has the wrong sign, even more so than for the benchmark Fama-MacBeth estimates. As discussed above, these full-sample betas and liquidity levels exhibit more cross-sectional correlation, but nevertheless the results still support our benchmark findings.

In the Online Appendix F, we present a wide range of robustness checks on the asset pricing results presented here. First, we use alternative liquidity measures for corporate bonds: the imputed roundtrip cost measure of Feldhütter (2012) and Dick-Nielsen, Feldhütter, and Lando (2012), the Roll measure, and a measure based on the frequency of trades in the spirit of Bekaert, Harvey, and Lundblad (2007). Second, we use a wide range of additional risk factors in the model: corporate bond market returns, government bond market returns, a government bond liquidity factor, and a volatility factor. Fourth, we add various

¹³ As the betas and the expected costs contain estimation noise, the standard errors of this regression are calculated using an extension of the method by Shanken (1992). The Internet Appendix of Bongaerts, de Jong, and Driessen (2011) provides more details on the procedure. This procedure uses a Newey and West (1987) correction to adjust for autocorrelation in the time series of forward-looking expected returns, and it also corrects for cross-sectional correlation between the expected returns across portfolios.

bond characteristics to the model, including proxies for credit risk and the bond duration. Finally, we study various subsamples and assess the time variation in liquidity premia. Across all these robustness checks, we continue to find that both the liquidity level and equity market liquidity risk affect corporate bond expected returns, while exposure to corporate bond liquidity risk always has an economically negligible effect.

2.4 Fit of expected returns and the credit-spread puzzle

Table 3 presents the average of the R^2 values for the weekly cross-sectional regressions. In an average week, we thus explain 59% of the cross-sectional variation of the weekly forward-looking expected returns. This is substantial, as obviously the weekly forward-looking expected returns are estimates of the true expected returns and thus contain estimation noise. The model fit is not perfect, however, and this is also indicated by the constant term in Table 3, which is significant across all specifications, except for the full-sample estimates in specification (9). For this full-sample regression, the R^2 is also much higher, as in this case measurement error in the weekly expected returns is averaged out to some extent.

Figure 2 graphs the fitted values of the risk premium according to the benchmark specification (1). The graphs present the average across the three liquidity proxies per rating/EDF category and shows the portfolios with high liquidity and low liquidity risk, and the portfolios with low liquidity and high liquidity risk. The graphs confirm that the liquidity level premium, together with risk premia for equity market and equity liquidity risk, explains most of the observed expected returns, with a negligible effect of corporate bond liquidity risk. Hence we conclude that the model provides a quite decent fit of expected corporate bond returns.

These results also shed some light on the credit spread puzzle. Huang and Huang (2012) show that structural models of default risk generate credit spreads well below observed credit spread levels. Figure 2 shows that incorporating liquidity effects, mainly the liquidity level and equity market liquidity risk, helps substantially to improve the fit of expected returns (and thus credit spreads). The positive estimate for the intercept does show that the model still underestimates the expected return on bonds; hence liquidity does not fully resolve the credit spread puzzle.

An important caveat for this discussion of the credit spread puzzle is that our coefficients (liquidity and risk premia) are fitted to match the level and cross-section of expected returns. The key question then is whether our estimates of liquidity and risk premia are “reasonable” and in line with theoretical models. The equity premium estimate looks reasonable (5.03% per year), and the equity liquidity risk premium is of similar size, which is predicted by the model of Acharya and Pedersen (2005). Finally, the coefficient on the liquidity level can be related to the trading frequency of bonds under some assumptions. Specifically, Amihud and Mendelson (1986) show that

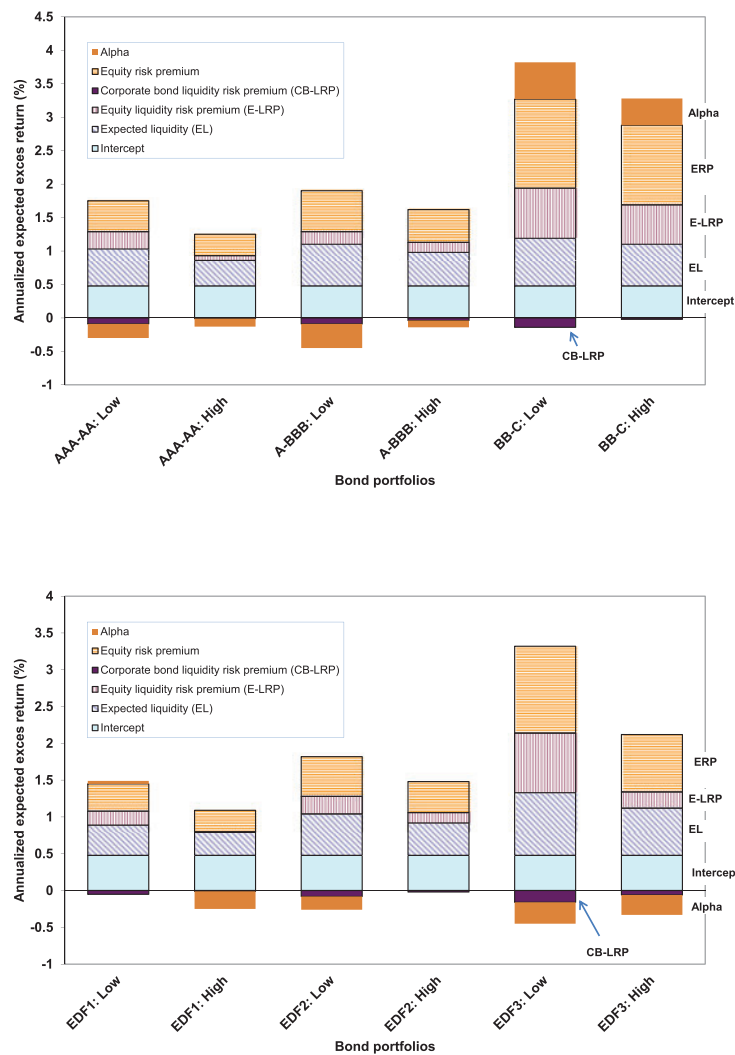


Figure 2
Fit of expected returns across portfolios

The figure shows the fitted values of the expected bond returns, obtained by multiplying the estimated coefficients in Table 3, specification (1) with the estimated transaction costs and the estimated betas. Alpha is the pricing error as defined in Equation (2). The fit is presented for portfolios across credit quality (rating categories/EDFs). For each credit quality level, the graph presents the fit for portfolios with low liquidity and highly negative liquidity betas and for portfolios with high liquidity and low liquidity risk (liquidity betas close to zero), averaged across the three liquidity proxy sorts (amount issued, age, and activity).

if investors are homogenous in their trading frequency, the coefficient on the roundtrip transaction costs directly equals the trading frequency. In our empirical model, the coefficient of 1.06 (for annualized expected returns) in specification (1) multiplies the one-way transaction costs c . The coefficient

for roundtrip transaction costs would thus be $1.06/2=0.53$. This would then correspond to a trading frequency of about twenty-three months ($1/0.53 \times 12$), or a turnover rate of 53% per year. This implied turnover rate is very close to the observed value-weighted turnover rate for the corporate bond market, which equals 55.2% per year (see Section 3). Hence, our liquidity level premium is in line with the prediction of a simplified Amihud-Mendelson model with homogenous trading frequencies.

2.5 Measurement error and multicollinearity in betas and transaction costs

A key concern for our benchmark regressions is that the variables on the right-hand-side, betas and transaction costs, are estimated and thus exhibit estimation error. This may bias our estimates. In particular, if the measurement error in CBLIQ betas is much larger than the measurement error in transaction cost estimates, our analysis may bias toward finding a liquidity level effect and against finding a significant corporate bond liquidity risk premium. An additional concern is multicollinearity. As shown in Table 1, the variables in our cross-sectional regressions are correlated and it is fair to ask how this affects the estimation and inference. To analyze these two issues in detail, we perform a simulation study that mimics as much as possible our empirical setup. We describe the most important aspects of our simulation study here. In Appendix D we describe the setup in full detail.

The first part of the simulation setup is a standard linear asset pricing model for dependent variable $y_{i,t}$, which gives the “true” expected return on portfolio i in week t ,

$$y_{i,t} = \lambda_c c_i + \lambda_{CBLIQ} \beta_{CBLIQ,i} + \lambda_{EQLIQ} \beta_{EQLIQ,i} + \lambda_{EQ} \beta_{EQ,i} + \delta_i + \varepsilon_{i,t} \quad (8)$$

with four explanatory variables: transaction costs c_i and the three betas, which are all assumed to be constant over time. We also add two types of pricing errors. First, a variable δ_i , which is independent across portfolios, has mean zero and a normal distribution with variance σ_δ^2 . This term captures pricing errors that do not disappear asymptotically. Second, we incorporate a pricing error term $\varepsilon_{i,t}$ with mean zero and normal distribution with variance σ_ε^2 , and also independent across portfolios, but this term is time-specific and follows an AR(1) process over time. We include these two types of pricing errors to fit both the full-sample and Fama-MacBeth R^2 values.

To obtain a cross-section of true betas and costs, we assume

$$\begin{pmatrix} c_i \\ \beta_{CBLIQ,i} \\ \beta_{EQLIQ,i} \\ \beta_{EQ,i} \end{pmatrix} \sim N(\mu_{c\beta}, \Sigma_{c\beta}) \quad (9)$$

where $\Sigma_{c\beta}$ is a 4×4 covariance matrix capturing the cross-sectional variances and correlations of the explanatory variables across the portfolios, and $\mu_{c\beta}$ is a

4×1 vector that captures the average costs and betas. In sum, the “true” model generates expected returns that are determined by the liquidity level and risk premia, which are all constant over time, and idiosyncratic pricing errors.

The second part of the setup introduces measurement error. We assume that all variables are estimated with error in each week, so that their observed values equal the true values plus measurement error

$$\widehat{c}_{i,t} = c_i + u_{c,i,t}, \quad (10)$$

$$\widehat{\beta}_{CBLIQ,i,t} = \beta_{CBLIQ,i} + u_{CBLIQ,i,t}, \quad (11)$$

$$\widehat{\beta}_{EQLIQ,i,t} = \beta_{EQLIQ,i} + u_{EQLIQ,i,t}, \quad (12)$$

$$\widehat{\beta}_{EQ,i,t} = \beta_{EQ,i} + u_{EQ,i,t}, \quad (13)$$

where all u -variables are measurement errors that are normally distributed with unconditional mean zero and independent across portfolios. As discussed in Appendix D in detail, we allow for persistence in these measurement errors by assuming AR(1) processes for each error. For each variable, we assume the variance of the measurement error to be constant across portfolios. For example, we have $u_{CBLIQ,i,t} \sim N(0, \sigma_{u,CBLIQ}^2)$.

To replicate one full Fama-MacBeth estimation, we (i) draw true values for costs and betas, (ii) simulate time series for the pricing errors, with a time series length of T weeks ($T=549$ to match the empirical sample size), (iii) calculate true expected returns $y_{i,t}$ using Equation (8) for each simulated week ($t=1, \dots, T$), (iv) draw a time series of measurement errors u , (v) construct observed values for the costs and betas by adding the measurement error to the true values, (vi) for each simulated week ($t=1, \dots, T$), perform a cross-sectional regression of $\widehat{y}_{i,t}$ on a constant and the observed values for costs $\widehat{c}_{i,t}$ and betas $\widehat{\beta}_{CBLIQ,i,t}$, $\widehat{\beta}_{EQLIQ,i,t}$, and $\widehat{\beta}_{EQ,i,t}$, and (vii) finally calculate the Fama-MacBeth estimates as the average of the weekly regression estimates, and calculate the t -statistic of this average using Newey-West with 24 lags. We repeat this entire scheme 5,000 times, which allows us to calculate the bias due to measurement error and the distribution of Fama-MacBeth estimates across simulations. We also calculate full-sample estimates in each simulation, which are obtained simply by first averaging all expected returns, costs, and betas over all 549 weeks, and then performing a single cross-sectional regression of the time series average of expected returns on the time series average of costs and betas.

Of course, the bias in the parameter estimates depends on the variance of the measurement errors, and the calibration of these variances is thus important. In general, the goal of the calibration is to match the cross-sectional and time series characteristics of the dependent and independent variables. Recall that for each portfolio we observe a time series of transaction costs and betas (obtained using the two-sided kernel). We then make the conservative assumption that the true transaction costs and betas are constant over time. This allows us to calibrate

the amount of measurement error since in this case all time series variation in the estimated costs and betas is due to measurement error. This assumption is conservative because in reality it is likely that the “true” levels of transaction costs and betas vary over time, so that some of the time series variation is not due to measurement error.

In Appendix D we describe in detail how we calibrate the parameters in the simulation setup. To identify the measurement error variances, we calibrate to two important statistics. The first is the time series average of the cross-sectional standard deviation of the weekly transaction costs (or weekly betas), which we denote Std_{FB} in panel A of Table 4. The second statistic is the cross-sectional standard deviation of the time series average of transaction costs (or betas), denoted by Std_{FS} . Due to averaging out of measurement error, the latter standard deviation should be smaller, which is exactly what we see in the data (Table 4, panel A). We can thus infer the amount of measurement error from the difference between these two standard deviations. In Appendix D we work this out mathematically. As shown in Table 4 (panel B), this calibration approach leads to substantial measurement error variances. For the transaction costs, corporate bond liquidity betas and equity liquidity betas,

Table 4
Simulation study: Calibration input and parameters

Panel A: Calibration inputs

	c	β_{CBLIQ}	β_{EQLIQ}	β_{EQ}
μ_X	0.517	-0.032	-0.067	0.131
λ_X	1.06	-3.59	-4.46	5.03
$Std_{FB}(\tilde{X})$	0.1981	0.0413	0.0766	0.0864
$Std_{FS}(\tilde{X})$	0.1527	0.0310	0.0582	0.0855

Panel B: Calibration results

	c	β_{CBLIQ}	β_{EQLIQ}	β_{EQ}
σ_X	0.147	0.027	0.055	0.085
$\sigma_{u,X}$	0.132	0.031	0.054	0.014
ρ_X	0.963	0.986	0.975	0.982

Panel C: True correlations

	c	β_{CBLIQ}	β_{EQLIQ}	β_{EQ}
c	1.00	-0.67	-0.65	0.77
β_{CBLIQ}	-0.67	1.00	0.60	-0.57
β_{EQLIQ}	-0.65	0.60	1.00	-0.96
β_{EQ}	0.77	-0.57	-0.96	1.00

Panel D: Calibration results: Pricing error parameters

	σ_δ	σ_ϵ	ρ_ϵ
	0.330	0.620	0.927

This table presents the different calibration inputs and outcomes used for the simulation exercise. Panel A provides calibration inputs. Panels B and C provide the calibration results of our calibration exercise as described in Section 2.5. Here σ_X is the true cross-sectional standard deviation of each of the four variables ($\sqrt{\text{diag}(\Sigma_{C\beta})}$). Panel D contains the calibrated values for the error term variables. Appendix C discusses the parameters and calibration approach in detail.

about half of the observed cross-sectional standard deviation is assigned to measurement error variation, with the remaining half representing true cross-sectional variation. Only for the equity beta the measurement error variance is much smaller. Appendix D also details how we fit the means and covariances of transaction costs and betas, the persistence of these variables over time, and the cross-sectional R^2 values reported in Table 3. In particular, we estimate the cross-sectional correlations of the true costs and betas. To be conservative, we estimate these using the full-sample cross-sectional correlations in panel C of Table 1, rather than the smaller Fama-MacBeth cross-sectional correlations (panel B of Table 1). The true correlations used in the simulations are then higher than the correlations of the estimated costs and betas due to presence of measurement error (panel C of Table 1 vs. panel C of Table 4). Our analysis thus builds in a substantial and conservative amount of correlation across the variables.

In Table 5 we present the results of our simulation study. We consider two sets of values for the (risk) premium parameters λ . In the first setting in panel A, we choose the λ -coefficients from the benchmark specification (1) in Table 3, except for the corporate bond liquidity risk premium, which has the wrong sign according to the empirical estimates. To see if we would identify a correctly signed corporate bond liquidity risk premium, we use the coefficient in specification (6) of Table 3. In this case $\lambda_{CBLIQ} = -3.59$, which implies a positive contribution of the corporate bond liquidity risk premium to the expected return. We report both the Fama-MacBeth estimates and the full-sample estimates. We start by analyzing the power of our tests. In particular, we calculate in what fraction of the 5,000 simulations we cannot reject that the estimated corporate bond liquidity risk premium is equal to zero, given that the true value for this premium is nonzero. We thus ask the question: “What is the probability that we do not reject the null that corporate bond liquidity is not priced ($H_0: \lambda_{CBLIQ} = 0$) when the true value of λ_{CBLIQ} is nonzero?”¹⁴ The table shows that, for our choice of parameters with $\lambda_{CBLIQ} = -3.59$, this probability is only 4.2% when we use a confidence level of 95%. Hence, we have substantial power to detect the corporate bond liquidity risk premium, if it exists.

Next we assess the biases in the coefficient estimates. For the Fama-MacBeth estimates, we find substantial biases for all variables. The liquidity level premium, corporate bond liquidity risk premium, and equity liquidity risk premium are all biased towards zero, while the equity premium is biased upwards. However, all coefficients have the correct sign and are on average statistically significant as indicated by the average of the t -statistics across simulations. Hence, even with these conservative levels of measurement error variance and cross-sectional correlations, we can detect the various (risk)

¹⁴ We thank the editor for suggesting this calculation.

Table 5
Simulation results with measurement error

λ_c	λ_{CBLIQ}	λ_{EQLIQ}	λ_{EQ}	const.	R^2
Panel A: Benchmark calibration					
True parameter values					
1.06	−3.59	−4.46	5.03	0	
Fama-McBeth estimates					
0.388	−2.103	−0.567	8.37	0.22	0.59
[5.052]	[−6.276]	[−2.474]	[37.302]	[6.517]	
{0.074}	{0.042}	{0.343}	{0}	{0.056}	
Full sample estimates					
0.784	−3.29	−1.759	7.079	0.067	0.848
[1.678]	[−1.884]	[−0.978]	[4.932]	[0.393]	
{0.492}	{0.41}	{0.739}	{0.002}	{0.886}	
Panel B: Zero liquidity level premium: $\lambda_c=0$					
True parameter values					
0	−3.59	−4.46	5.03	0	
Fama-McBeth estimates					
0.044	−1.614	−0.707	7.443	−0.023	0.519
[0.584]	[−4.855]	[−3.111]	[33.493]	[−0.675]	
{0.682}	{0.118}	{0.222}	{0}	{0.784}	
Full sample estimates					
−0.071	−2.986	−2.02	6.704	0.002	0.804
[−0.148]	[−1.718]	[−1.129]	[4.691]	[0.009]	
{0.956}	{0.479}	{0.69}	{0.004}	{0.944}	

The table contains the results for the simulation study described in Section 2.5, where we assess how measurement error in the liquidity level and betas affects the estimates for the premia on the liquidity level and liquidity and market risk. Panel A presents the benchmark results for a setup that mimics the benchmark Fama-MacBeth regressions and the full-sample regressions, respectively, using the parameters presented in Table 4. Panel B is similar, but we set the true value of the liquidity level premium equal to zero ($\lambda_c=0$). We perform 5,000 simulations. We report the averages across these simulations for the following variables: (i) the estimated liquidity level and liquidity and market risk premia, (ii) the t -statistics of these estimates (in square brackets), (iii) the cross-sectional R^2 , and (iv) the fraction of simulations for which we can not reject the null hypothesis of a coefficient equal to zero at a 95% confidence level (in curly brackets).

premia due to the sample size and the variation generated by sorting on both liquidity level and risk. In particular, the biases do not lead to a wrongly signed corporate bond liquidity risk premium. Table 5 also shows that especially the equity liquidity risk premium is biased downwards and the equity premium is biased upwards. This is because these two variables are strongly correlated, with smaller measurement error in equity betas.

Turning to the full-sample estimates, we find that the biases are smaller, which is as expected since measurement error is averaged out to some extent in this case. However, the full-sample regression uses less information than the weekly Fama-MacBeth regressions, and hence the t -statistics are smaller in this case. As a result, the “nonrejection” fraction equals 41.0% for λ_{CBLIQ} . This illustrates again that using the time variation in the liquidity level and factor betas is important to identify the various (risk) premia. For the full-sample estimates we also see that the liquidity level premium and corporate bond liquidity risk premium are still marginally significant (averaged across simulations), and the

equity liquidity risk premium becomes insignificant. Again, we do not find that we obtain the wrong sign for the corporate bond liquidity risk premium. In fact, the downward bias for this premium is only 8% (from the true value of -3.59 to -3.29).

We then turn to the second case for the λ -parameters, where we set the true liquidity level premium equal to zero ($\lambda_c=0$), keeping the other parameters at the same value. This is to see whether one would mistakenly conclude that the liquidity level matters, while it should be liquidity risk that matters. Panel B of Table 5 shows that this does not happen: the estimate for the liquidity level effect is close to zero and insignificant, while the corporate bond liquidity risk premium has the correct sign and is significant (strongly in the Fama-MacBeth case, marginally in the full-sample case). For the Fama-MacBeth estimation, the “non-rejection” fraction equals 11.8% for λ_{CBLIQ} . In other words, in 88.2% of the simulations do we find a significant corporate bond liquidity risk premium.

Finally, we push the impact of measurement error even further, and we consider various ratios of measurement error variance to total cross-sectional variance.¹⁵ Here we follow the calibration approach described above to calibrate all parameters, but then change how the total observed cross-sectional variance is divided between measurement error variance and “true” cross-sectional variance. Increasing the measurement error variance thus implies lower true cross-sectional variance. We change the ratio of measurement error variance to total variance for the liquidity level and corporate bond liquidity risk and consider levels of 0.25 and 0.75 (recall that our calibrated levels of the ratios are close to 0.5). Table 6 contains the results, where we focus on the case where the true effect of the liquidity level is zero ($\lambda_c=0$), to see if we would mistakenly conclude that the liquidity level matters due to measurement error. The results show that this does not happen: the liquidity level premium estimate is close to zero and insignificant. The most interesting case is perhaps panel E, where we increase the ratio for the corporate bond liquidity beta to 0.75 and decrease the ratio for the liquidity level to 0.25. In this case, the liquidity level is measured much more precisely, which could bias toward finding a liquidity level effect, and against finding a liquidity beta effect. Panel E shows, however, that these biases are negligible for the liquidity level effect. There is a downward bias for the corporate bond liquidity risk premium, but it still has the right sign (and is significant in the Fama-MacBeth case). Hence based on these results we think it is unlikely that measurement error can explain why we empirically find the wrong sign for the corporate bond liquidity risk premium. In sum, Panels A-E show that varying the ratios does affect the biases, as expected, but the effects seem not large enough to support the conclusion that measurement error explains our empirical findings.

¹⁵ We thank an anonymous referee for this suggestion.

Table 6
Simulation results: Ratios approach

λ_c	λ_{CBLIQ}	λ_{EQLIQ}	λ_{EQ}	const	R^2
True parameter values					
0	-3.59	-4.46	5.03	0	
Panel A: $\sigma_{u,c}^2/\sigma_c^2=0.4469$; $\sigma_{u,\beta}^2/\sigma_\beta^2=0.25$					
Fama-McBeth					
-0.0019 [-0.023]	-3.0199 [-8.752]	-0.6582 [-2.895]	7.3138 [32.663]	-0.0234 [-0.687]	0.5348
Full sample					
-0.1736 [-0.36]	-3.8631 [-2.504]	-1.7809 [-0.989]	6.8118 [4.775]	0.0287 [0.163]	0.8146
Panel B: $\sigma_{u,c}^2/\sigma_c^2=0.4469$; $\sigma_{u,\beta}^2/\sigma_\beta^2=0.75$					
Fama-McBeth					
0.0457 [0.61]	-0.9302 [-2.864]	-0.7095 [-3.129]	7.4464 [33.72]	-0.002 [-0.062]	0.5097
Full sample					
-0.0209 [-0.042]	-2.169 [-1.149]	-2.1385 [-1.206]	6.6502 [4.666]	0.0011 [0.005]	0.7967
Panel C: $\sigma_{u,c}^2/\sigma_c^2=0.25$; $\sigma_{u,\beta}^2/\sigma_\beta^2=0.5643$					
Fama-McBeth					
0.0631 [0.765]	-1.5942 [-4.75]	-0.7135 [-3.13]	7.4053 [31.432]	-0.0271 [-0.815]	0.5193
Full sample					
-0.0694 [-0.159]	-3.0003 [-1.709]	-2.0085 [-1.114]	6.7212 [4.601]	-0.001 [-0.008]	0.8039
Panel D: $\sigma_{u,c}^2/\sigma_c^2=0.75$; $\sigma_{u,\beta}^2/\sigma_\beta^2=0.5643$					
Fama-McBeth					
0.0233 [0.341]	-1.634 [-4.965]	-0.7021 [-3.097]	7.479 [36.086]	-0.0166 [-0.479]	0.5193
Full sample					
-0.0672 [-0.116]	-2.9471 [-1.743]	-2.0516 [-1.168]	6.657 [4.97]	0.0051 [0.02]	0.8038
Panel E: $\sigma_{u,c}^2/\sigma_c^2=0.25$; $\sigma_{u,\beta}^2/\sigma_\beta^2=0.75$					
Fama-McBeth					
0.0645 [0.795]	-0.9161 [-2.805]	-0.715 [-3.146]	7.4075 [31.548]	-0.0064 [-0.199]	0.5098
Full sample					
-0.0207 [-0.046]	-2.1737 [-1.144]	-2.135 [-1.197]	6.6559 [4.574]	0.0003 [0.001]	0.7968

This table presents Fama-McBeth and full-sample regression results for our simulation study where we keep the calibrated correlations, pricing error volatilities, and autocorrelation parameters at the levels given in Table 4, but vary the fraction of measurement error variance in, respectively, transaction costs and corporate bond liquidity betas, as a fraction of the total observed cross-sectional variance of this variable. The measurement error variance of equity market risk and equity liquidity risk is kept at the level reported in Table 4. In these simulations, the liquidity level premium λ_c is set to zero.

In Online Appendix F, we present results for various alternative parameter settings. First, we set the equity risk premium to zero, and find that this does not affect the inference on the bond liquidity level and risk. We do see that, because of the correlation between equity liquidity and equity market risk, the equity premium is biased upwards in this parameter setting. Second, we set the corporate bond liquidity risk premium to zero to obtain a setting that is close to our empirical estimates. We find that the liquidity level premium is biased toward zero, while in only 33.2% of the simulations we find an insignificant corporate bond liquidity risk premium, even though the true value is zero. Hence, in this case there is a bias toward finding a liquidity risk premium and against finding a liquidity level premium. Finally we use a lower value for the corporate bond liquidity risk premium, which implies an average contribution of six basis points to the expected return. Even with such a small economic effect, the nonrejection fraction is only 13.5% for the Fama-MacBeth estimation. In Online Appendix F, we also graph the simulation estimates across simulations to verify that the parameter distribution does not contain outliers and is not skewed or fat-tailed.

In sum, this section shows that even with extreme assumptions on measurement error in the liquidity beta and conservative levels for the cross-sectional correlations, one would still detect a corporate bond liquidity risk premium if it were to exist. Hence it seems unlikely that measurement errors or multicollinearity explain our main empirical findings.

3. Why Is Corporate Bond Liquidity Risk Not Priced?

Our results show that the liquidity level and equity market liquidity risk have a strong effect on corporate bond prices, while corporate bond liquidity risk exposure does not. We now provide a potential explanation for these findings. The starting point of our theoretical model is that, if an investor experiences a funding liquidity shock and is forced to sell assets, she will prefer to sell liquid assets with low transaction costs and hold on to illiquid assets. Below we show what this implies for liquidity risk premia.

In this setup, corporate bonds serve as the illiquid asset. For the liquid asset, one could consider equity or Treasury bonds. We focus on equity as the liquid asset, and not Treasury bonds, for two reasons. First, the prime holders of corporate bonds are insurance companies and pension funds (see Campbell and Taksler 2003). These institutions in general have long-term claims on their balance sheets with sizable interest rate exposure. To this end, these institutions hold long maturity Treasuries to hedge their interest rate risk. Selling these bonds would compromise their hedging strategy. Hence, it might be more likely that these investors sell equity and not Treasury bonds in crisis periods. Second, in our empirical analysis we focus on corporate bond returns in excess of the return on a Treasury bond with the same maturity. This implies that factors driving Treasury bond returns, including Treasury bond liquidity risk, are not

directly relevant for excess corporate bond returns. At the end of this section, we discuss this in more detail.

We thus consider an investor holding both liquid assets (U.S. equities) and illiquid assets (U.S. corporate bonds).¹⁶ There are various reasons why institutional investors trade these assets over time, such as rebalancing, risk-shifting, satisfying regulatory capital requirements, exogenous liquidity needs, etc. In the presence of transaction costs, the investor then faces a trade-off between having an optimally diversified portfolio and minimizing trading costs (see for example Constantinides 1986).

In Online Appendix E, we formalize such a setup using a simple two-period asset pricing model. In this model, a mean-variance investor can invest in two assets, a low-cost asset and high-cost asset. After one period, the investor is forced to liquidate part of his portfolio due to an exogenous liquidity shock. If the difference between transaction costs of the two assets is sufficiently high, it is optimal for the investor to absorb the liquidity shock by selling only low-cost assets and avoid trading in the high-cost asset. In equilibrium, this implies that the risk of innovations in the liquidity of the high-cost asset is not priced. Only exposure to the liquidity risk of the low-cost asset is priced. Applying this result to our setting, it would imply that corporate bond liquidity risk is not priced, which is in line with our empirical findings, while exposure to equity market liquidity risk should be priced. This is also in line with our empirical findings: in Table 3 we find that exposure to equity market illiquidity has a significant risk premium with the expected sign. Alternatively, if the difference between transaction costs of the two assets is sufficiently small, our simple model predicts that both assets are used to absorb the liquidity shock, although the low-cost asset is still used relatively more. In this case, innovations in transaction costs of both assets are priced.

This simple model can be used to generate two additional testable hypotheses. First, the model predicts lower turnover for the high-cost asset. This prediction is obviously not unique to our theoretical model because most models of liquidity and trading predict less trading when the asset is less liquid. Second, the pricing of liquidity and the turnover levels may be different in normal times versus crisis periods. To see this, consider the case where, in normal times, transaction costs on both assets are sufficiently small so that both assets will be used for trading. In this case, liquidity risk on both the low-cost asset (equity) and high-cost asset (corporate bonds in our case) is priced. However, in normal times the variation in transaction costs is quite small (see Figure 1), and hence the effect of liquidity risk on prices may be negligible. Next, consider a crisis period during which transaction costs on the high-cost asset increase

¹⁶ Note that U.S. equities are typically much more liquid than U.S. corporate bonds. Hasbrouck (2009) has estimated transaction costs for U.S. equities using Roll's measure. Using his data, we find that for the S&P 500 stocks the average one-way cost is 0.15% in 2005–07, and 0.32% in 2008. In contrast, Figure 1 reports market-wide corporate bond transaction costs around 0.5% on average, increasing to about 1% during the crisis.

substantially so that in equilibrium this asset will not be used to absorb liquidity shocks. The model then predicts that trading in the liquid asset actually increases when moving from normal times to crisis periods, while trading in the illiquid asset decreases. This second prediction is much harder to obtain in other models. For example, standard models of trading and transaction costs (in the spirit of Constantinides 1986) always predict less trading when transaction costs are higher.

To test these predictions, we study turnover patterns in equity and corporate bond markets. For our 2003–13 sample period, we calculate monthly turnover for each stock in the CRSP database and each bond in our TRACE sample. We first focus on the value-weighted average turnover levels in both markets. Not surprisingly, we find that equity market turnover is much higher than corporate bond turnover. For stocks, the value-weighted average turnover level equals 65.7% per quarter, while for corporate bonds the corresponding number is 13.8% per quarter.

Subsequently, we study the dynamics of turnover. We perform two sets of regressions. First, we regress monthly changes in the value-weighted average equity turnover on the equity market return, the change in the market-wide equity ILLIQ measure, and the change in the market-wide level of corporate bond transaction costs. Results in Table 7 show that a decrease in equity liquidity or a decrease in the stock market index level implies a significant increase in equity market turnover. Corporate bond liquidity does not affect equity turnover. These results show that equity market turnover increases in bad, illiquid times. As discussed above, this may be because investors are essentially forced to trade despite the higher transaction costs.

We then perform a similar regression for market-wide corporate bond turnover, which we regress on the change in the market-wide level of corporate

Table 7
Equity and corporate bond turnover regressions

	(1) <i>dbondto</i>	(2) <i>deqto</i>	(3) <i>dbondto</i>	(4) <i>deqto</i>
<i>dbondliq</i>	-1.138 [-4.61]	-0.106 [-0.49]	-1.135 [-4.57]	-1.770 [-0.82]
<i>deqliq</i>	-0.011 [-0.22]	0.167 [3.70]	0.001 [0.01]	0.118 [2.32]
<i>mktrf</i>				-0.805 [-2.04]
<i>dcs</i>			-1.001 [-0.22]	
<i>N</i>	124	124	124	124
<i>R</i> ²	0.152	0.101	0.152	0.131

The table contains estimates of linear regressions of monthly changes in the value-weighted mean of log monthly turnover across corporate bonds (*dbondto*, specification (1) and (3)) or stocks (*deqto*, specification (2) and (4)) on market-wide corporate bond illiquidity changes (*dbondliq*), changes in the value-weighted, inflation-corrected ILLIQ across stocks (*deqliq*), changes in value-weighted average credit spreads (*dcs*), and the value-weighted equity market excess return (*mktrf*). Data runs from July 2003 to December 2013; the data point for July 2012 is excluded due to the switch from TRACE Enhanced to TRACE as data source and the associated imposed volume caps. *t*-statistics are in brackets.

bond transaction costs, the change in the market-wide equity ILLIQ measure, and the change in the average credit spread across all bonds. The results in Table 7 show that corporate bond turnover decreases when the market becomes less liquid and when credit spreads increase (though the latter effect is not significant). Hence, in illiquid times, corporate bond turnover decreases.

In sum, these results shows that stocks have much higher turnover than corporate bonds, and that this difference is larger when prices go down and when markets are overall less liquid. Hence, in line with our hypothesis, when transaction costs increase, investors prefer to trade the relatively more liquid assets and avoid trading in less liquid assets.

Finally, as discussed above, one could also have a Treasury bond as the liquid asset. If this is the case, Treasury market liquidity risk should be priced in the cross-section of corporate bond returns. In Online Appendix F, we use the Treasury market liquidity measure of Hu, Pan, and Wang (2013) and do not find evidence that Treasury market liquidity risk is priced. This can be understood from the fact that we focus on explaining expected corporate bond returns in excess of maturity-matched Treasury bond returns. In Online Appendix F, we work out what the model predicts for these excess corporate bond returns if the two risky assets are a corporate bond and a Treasury bond. This analysis shows that in this case only the covariance of the excess corporate bond return and Treasury liquidity risk is priced. Given the focus on the excess return, this covariance is likely small. We confirm this empirically. When we scale the Treasury liquidity risk factor to have the same variance as equity market returns, we find that the univariate beta of excess corporate bond returns with respect to Treasury market liquidity shocks is small: averaged across portfolios, it equals -0.05 . Of course, this does not mean that there is no Treasury liquidity risk premium. But if it exists, it is of negligible importance for excess corporate bond returns due to the small beta with respect to Treasury liquidity shocks.

4. Conclusion

This paper studies the asset pricing implications of the liquidity level and liquidity risk for expected corporate bond returns. We measure corporate bond liquidity at the portfolio level by the effective transaction costs using a repeat-sales procedure to signed transaction prices. We then construct liquidity levels and liquidity innovations for a set of corporate bond portfolios. Several asset pricing models are estimated using the cross-section of corporate bond portfolios. Overall, we find a strong effect of the liquidity level and equity market liquidity risk on expected corporate bond returns, while there is little evidence that corporate bond liquidity risk exposures explain expected corporate bond returns, even during the recent financial crisis. We perform a wide variety of robustness checks to validate this result, and using a simulation study we argue that it is unlikely that measurement error in betas or multicollinearity can explain our results. Our findings are consistent with

a setting in which investors strategically choose to use the assets with low liquidity costs to rebalance their portfolio and accommodate other trading needs.

Supplementary Data

Supplementary materials for this article are available online at <http://rfs.oxfordjournals.org/>.

Appendix A. Data Filters

For our study we use a combination of the TRACE Enhanced and the regular TRACE databases. The TRACE Enhanced database contains all reported trades, transaction sizes, and trade sign indicators since the inception of July 2002. The regular TRACE database contains the same data, but has only partial coverage from July 2002 to October 2004, has transaction sizes capped at \$1 million for high-yield bonds and \$5 million for investment grade bonds and only starts to distribute trade sign indicators from October 2008 onwards. Hence, one would strictly prefer to use TRACE Enhanced over the regular TRACE data. However, the disclosure of TRACE Enhanced is delayed by 1.5 years. For that reason, we use TRACE Enhanced from July 2002 to June 2012, and regular TRACE from July 2012 to December 2013.

We apply several filters to our data set to remove bonds with special features and to remove erroneous entries. Our filters are very similar to those employed in Bongaerts, Cremers, and Goetzmann (2012). We remove all trades that include commission, that have a settlement period of more than five days, “when-issued” trades, trades with special conditions, and all trades that are canceled or reversed. For July 2012 to December 2013, trade volumes are truncated by the system and we replace truncated trade volumes by their respective truncation barrier (\$5 million for investment grade and \$1 million for high yield). We remove all trades for which we have a negative reported yield, since these will be mainly driven by implicit option premia in the yield. We use Mergent FISD to match the trades to bond characteristics and S&P ratings using CUSIPs. We discard all bonds with convertibility options, that are puttable, that have a nonfixed coupon, that are subordinated, secured or guaranteed, perpetual or in foreign currency. We keep callable bonds because they comprise a large part of the sample. Removing these callable bonds would substantially reduce the sample size and the precision of our transaction costs estimates. We discard all zero-coupon bonds. We also remove trades with a settlement date later than or equal to the maturity date. Furthermore, we found several duplicate records, resulting from both parties involved in a trade reporting to the system. We filter out these trades by consecutively sorting on bond, date, and volume and removing identical consecutive records. Moreover some of the yield changes are unrealistically high. Therefore, we remove trades with yield changes of more than 1,000 basis points. We also exclude all trades with a volume lower than \$10,000 since very small trades may have substantially higher transaction costs. As Ford and GM together were responsible for more than 10% of all corporate bond trading up to the default of GM in 2009, we excluded these issuers from the benchmark analysis to avoid portfolios to be completely driven by individual companies. In Online Appendix F, we show that adding a separate portfolio with the Ford and GM bonds to the cross-section does not affect our findings. Finally, we remove interdealer trades. We thus focus on buyer- versus seller-initiated transactions as these allow us to directly measure effective transaction costs.

Appendix B. Triple Sorts

In this Appendix we provide details about the portfolio sorts. To perform the triple sort on credit quality, liquidity proxies, and liquidity beta, we need an estimate of the liquidity beta of each bond.

We now describe the procedure to calculate these. In short, the liquidity beta for each bond is calculated as a weighted average of the direct beta estimate at the bond level, and the portfolio betas for double-sorted portfolios in which this bond is present.

We first perform double sorts (see also Online Appendix F.1), where we sort on credit quality (five categories) and a liquidity proxy (two categories), where we use the rating or EDF to measure credit quality, and age, amount issued or trading activity as liquidity proxy. We apply the repeat-sales model of Section 1.3 to estimate portfolio returns and transaction costs for all these portfolios. We then estimate portfolio liquidity betas and their standard errors from a regression of bond portfolio excess returns on market liquidity innovations for all double-sorted portfolios across quality (rating and EDF) and liquidity (age, issue size, trading volume) as well as the AAA- and CCC-rated portfolio. We do this on a weekly basis using a one-year rolling window. For each portfolio we then create quarterly betas and standard errors by averaging the betas over all weeks in the quarter and calculating the appropriate covariances. Next, we average for each bond-quarter the betas and squared standard errors¹⁷ across all portfolios in which that bond was contained.

For each bond, we also estimate the direct liquidity beta. To this end, we estimate a beta and standard error from the univariate regression of the individual bond excess returns on corporate bond market liquidity innovations. We again do this on a weekly basis using the last trade available every week on a one-year rolling window, where we require at least 25 observations and where the smallest and largest observation are winsorized.

When constructing the portfolios for a given quarter, say Q2 2006, the one-year rolling window used to estimate these betas includes this quarter; hence we use data from Q3-Q4 2005 and Q1-Q2 2006 in this example. We thus essentially use a mixture of the pre-ranking and post-ranking betas to form portfolios. This has the advantage that it generates more variation in the liquidity betas used to estimate the asset pricing model. The disadvantage is that these portfolios are not ex-ante tradable portfolios, but this is not an issue for estimating and testing asset pricing models.

Having obtained the portfolio beta and the direct beta, we can now use the standard Bayesian formula to calculate our posterior beta

$$\beta_{liq}^{post} = \frac{var(\hat{\beta}_{liq}^{port})^{-1} \hat{\beta}_{liq}^{port} + var(\hat{\beta}_{liq}^{direct})^{-1} \hat{\beta}_{liq}^{direct}}{var(\hat{\beta}_{liq}^{port})^{-1} + var(\hat{\beta}_{liq}^{direct})^{-1}}. \quad (B1)$$

For the triple sorts, we have three rating or EDF categories (to prevent an excessive number of portfolios with few bonds per portfolio). Within each category, we then first sort on a liquidity proxy (amount issued, age and activity; two portfolios) and then on the posterior liquidity beta $\hat{\beta}_{liq}^{post}$ discussed above (again two portfolios). In total, the triple sort thus gives $3 \times 2 \times 2$ portfolios for each credit quality measure (rating or EDF) and each liquidity proxy (age, amount issued, or trading activity).

Appendix C. The Repeated Sales Implementation

In this Appendix, we summarize the repeated sales methodology as in Case and Shiller (1987) and provide further detail on the extensions we add to incorporate and estimate transaction costs. We start with the simple case where there are no transaction costs. Consider a return process of the fundamental values $V_{j,t}$ of all assets j belonging to a homogeneous portfolio i in the absence of transaction costs

$$\ln(V_{j,t}) - \ln(V_{j,t-1}) = R_{i,t} + v_{j,t}, \quad (C1)$$

where $R_{i,t}$ is a one-period fundamental return of the portfolio and $v_{j,t}$ is an i.i.d. idiosyncratic return component. Hence, returns of individual asset returns consist of a common as well as idiosyncratic

¹⁷ This implicitly implies a correlation of one between beta estimates of different portfolios; betas are typically highly correlated across liquidity sorts. If anything, this would put too little weight on the portfolio betas.

component. Transactions, however, do not take place at regular intervals, but rather at intervals of random and unequal length. Consider two subsequent transactions in the same asset; one takes place at t_k and the other at t_{k-1} . We have

$$\begin{aligned} \ln(V_{j,t_k}) - \ln(V_{j,t_{k-1}}) &= \sum_{s=t_{k-1}+1}^{t_k} R_{i,s} + e_{j,t_k} \\ &= \sum_{s=1}^T R_{i,s} I_{s \in [t_{k-1}+1, t_k]} + e_{j,t_k}, \end{aligned} \quad (C2)$$

where T is the end of the sample, $e_{j,t_k} = \sum_{s=t_{k-1}+1}^{t_k} v_{j,s}$, and $I_{s \in [t_{k-1}+1, t_k]}$ is an indicator variable that equals 1 if $s \in [t_{k-1}+1, t_k]$ and 0 otherwise. Hence, the return between two transactions consists of $t_k - t_{k-1}$ one-period fundamental returns plus an idiosyncratic return. In the basic repeated sales methodology, one stacks subsequent (multi-period) returns based on observed consecutive transactions for all assets belonging to portfolio i . Next, one regresses these returns on dummy variables per time period, where the dummy variable equals 1 if the time period in question is contained in the interval between the two subsequent trades of that observation. The regression coefficients on these dummies are the estimates of the portfolio return $R_{i,t}$ in each period. Naturally, we can only estimate those when at each point in time we have at least one observation. Note that, since the portfolio returns $R_{i,t}$ are regression coefficients, we do not need any assumptions on the dynamics of these returns. As discussed below, we only need the usual assumptions on the error term e_{j,t_k} to obtain consistent estimates of the portfolio returns.

Now consider transaction prices in the presence of transaction costs. As discussed in Section 1.3, we have

$$P_{j,t_k} = V_{j,t_k} + c_i P_{j,t_k} Q_{j,t_k}, \text{ for all bonds } j \text{ in portfolio } i,$$

where $Q_{j,t}$ is the observed trade direction, which equals +1 in case of a buyer-initiated trade and -1 in case of a seller-initiated trade. We assume that all bonds j in portfolio i have the same transaction costs at a given point in time, and we thus denote c_i as the effective percentage half-spread. For two subsequent trades, the (multi-period) price return is given by (using a log approximation)

$$\begin{aligned} \ln(P_{j,t_k}) - \ln(P_{j,t_{k-1}}) &= \ln(V_{j,t_k}) - \ln(V_{j,t_{k-1}}) + c_i Q_{j,t_k} - c_i Q_{j,t_{k-1}} \\ &= \sum_{s=1}^T R_{i,s} I_{s \in [t_{k-1}+1, t_k]} + c_i Q_{j,t_k} - c_i Q_{j,t_{k-1}} + e_{j,t_k}. \end{aligned} \quad (C3)$$

We can stack these observations as before, and estimate $R_{i,s}$, $c_i \forall s$ by OLS. As before, the coefficient estimates on the time dummies $I_{s \in [t_{k-1}+1, t_k]}$ are estimates of the portfolio returns $R_{i,s}$ for each period s ; the coefficient estimate on the net trade sign indicators $Q_{j,t_k} - Q_{j,t_{k-1}}$ is the estimate of the portfolio transaction cost c_i .¹⁸

For our asset pricing regressions, we focus on weekly returns. However, it is important that we allow fundamental prices to fluctuate at a higher frequency to filter out fundamental price movements as much as possible, so that we can more precisely identify the transaction cost effects. Therefore, we let fundamental values fluctuate over a horizon of an hour, while we assume that transaction costs fluctuate at a weekly basis. We thus use an hourly frequency and estimate the repeat-sales model week by week.¹⁹

¹⁸ In practice, we pool the weekly models and estimate the weekly transaction costs and hourly returns at once for an entire quarter (the frequency at which we reform portfolios). This way we also include the price change from a transaction at the end of a given week to the next transaction in the following week.

¹⁹ We incorporate the exact timing of the trade by scaling the indicator functions. For example, if on a given day a first transaction occurs at 9.15 a.m. and a second transaction at 11.30 a.m., the price change has a loading of

Under which assumptions does this model provide consistent estimates of the effective transaction costs? As the model in Equation (6) is a standard regression model, we need the usual assumption that the error term $e_{j,t}$ is orthogonal to the explanatory variables. In our case, this means first of all that we assume that idiosyncratic shocks are orthogonal to the common component $R_{i,s}$. This is a standard market model assumption. Second, we assume that trade directions are orthogonal to the idiosyncratic shock. In economic terms, this means that transactions do not have permanent price impact. Again, this is a standard assumption for a Roll-style model of effective transaction costs. Importantly, we thus do not need to make assumptions on whether the common component or idiosyncratic shock is homoscedastic or heteroscedastic. Hence, the model allows for much larger shocks in credit risk during the crisis period. Also, in contrast to the Roll measure, which is based on the autocovariance and requires that fundamental price changes are uncorrelated over time, our repeat-sales model with trade directions does not require that the common factor or idiosyncratic component has zero autocorrelation. It is also important to note that, even if the stronger assumptions that are required for Roll's autocovariance measure would hold, our method is likely more efficient as it uses the observed trade signs to estimate effective transaction costs.

Appendix D. Simulation Study

In this Appendix we discuss the simulation setup in full detail and describe how we calibrate the parameters in this simulation study.

D.1 Simulation setup

Empirically, we perform weekly regressions using Fama-MacBeth. The first part of the simulation setup is therefore a standard linear asset pricing model for dependent variable $y_{i,t}$, which represents the expected return on portfolio i in week t ,

$$y_{i,t} = \lambda_c c_i + \lambda_{CBLIQ} \beta_{CBLIQ,i} + \lambda_{EQLIQ} \beta_{EQLIQ,i} + \lambda_{EQ} \beta_{EQ,i} + \delta_i + \varepsilon_{i,t} \quad (D1)$$

with four explanatory variables: transaction costs c_i and three betas, the corporate bond liquidity beta, equity liquidity beta, and equity market beta, which are all assumed to be constant over time. We also add two types of pricing errors. First, a variable δ_i , which is independent across portfolios, has mean zero and a normal distribution with variance σ_δ^2 . This term captures pricing errors that do not disappear asymptotically. Second, we incorporate a pricing error term $\varepsilon_{i,t}$ with mean zero and normal distribution with variance σ_ε^2 , also independent across portfolios, but this term is time-specific, persistent, and averages out over time. As discussed below, we include these two types of pricing errors to fit both the full-sample and Fama-MacBeth R^2 values.

To obtain a cross-section of true betas and costs, we assume

$$\begin{pmatrix} c_i \\ \beta_{CBLIQ,i} \\ \beta_{EQLIQ,i} \\ \beta_{EQ,i} \end{pmatrix} \sim N(\mu_{c\beta}, \Sigma_{c\beta}), \quad (D2)$$

where $\Sigma_{c\beta}$ is a 4×4 covariance matrix capturing the cross-sectional variances and correlations of the explanatory variables across the portfolios, and $\mu_{c\beta}$ is a 4×1 vector that captures the average costs and betas. In sum, the “true” model generates expected returns that are determined by the liquidity level and risk premia, which are all constant over time, and idiosyncratic pricing errors.

0.75 on the dummy for the 9 a.m.–10 a.m. period, a loading of 1 on the 10 a.m.–11 a.m. dummy, and a loading of 0.5 on the 11 a.m.–12 a.m. dummy for this day. This approach assumes that within each hour, the fundamental return is spread evenly over the minutes within this hour.

The second part of the setup introduces measurement error. We assume that all variables are estimated with error in each week, so that their observed values equal the true values plus measurement error

$$\widehat{c}_{i,t} = c_i + u_{c,i,t}, \quad (D3)$$

$$\widehat{\beta}_{CBLIQ,i,t} = \beta_{CBLIQ,i} + u_{CBLIQ,i,t}, \quad (D4)$$

$$\widehat{\beta}_{EQLIQ,i,t} = \beta_{EQLIQ,i} + u_{EQLIQ,i,t}, \quad (D5)$$

$$\widehat{\beta}_{EQ,i,t} = \beta_{EQ,i} + u_{EQ,i,t}, \quad (D6)$$

where all u -variables are measurement errors that are normally distributed with unconditional mean zero, and independent across portfolios. As discussed below, we allow for persistence in these measurement errors. For each variable, we assume the variance of the measurement error to be constant across portfolios. For example, we have $u_{CBLIQ,i,t} \sim N(0, \sigma_{u,CBLIQ}^2)$.

The final step of the simulation setup considers the time series variation in the measurement errors and pricing errors. This step is only important to generate a realistic distribution of simulated parameter estimates, and it does not affect the size of the bias due to measurement error. Empirically, we see that the transaction costs, betas, and pricing errors are persistent over time. This is not surprising for the betas, as they are estimated using a rolling window of data around a given week (using the two-sided kernel). It is thus possible that measurement errors are correlated over time. We incorporate this using an AR(1) specification for each measurement error variable

$$Corr(u_{c,i,t}, u_{c,i,t-k}) = \rho_c^k, \quad (D7)$$

$$Corr(u_{CBLIQ,i,t}, u_{CBLIQ,i,t-k}) = \rho_{CBLIQ}^k, \quad (D8)$$

$$Corr(u_{EQLIQ,i,t}, u_{EQLIQ,i,t-k}) = \rho_{EQLIQ}^k, \quad (D9)$$

$$Corr(u_{EQ,i,t}, u_{EQ,i,t-k}) = \rho_{EQ}^k, \quad (D10)$$

where the ρ -parameters capture the persistence. Similarly, we allow for persistence in the pricing error of the cross-sectional pricing equation (D1),

$$Corr(\varepsilon_{i,t}, \varepsilon_{i,t-k}) = \rho_\varepsilon^k. \quad (D11)$$

To replicate one full Fama-MacBeth estimation, we then perform the following steps:

1. Simulate true values for costs and betas, drawing from the multivariate normal distribution in Equation (D2), and simulate the “asymptotic” pricing errors δ_i .
2. For each portfolio and each variable (costs, betas), draw a time series of measurement errors u and pricing errors ε , correlated over time following equations (D7) to (D11) with a time series length of T weeks ($T=549$ empirically).
3. Construct observed values for the costs and betas by adding the measurement error to the true values.
4. Given the true costs, betas, and pricing errors, calculate true expected returns $y_{i,t}$ using Equation (D1) for each simulated week ($t=1, \dots, T$).
5. For each simulated week ($t=1, \dots, T$), perform a regression of $y_{i,t}$ on a constant and the observed values for costs $\widehat{c}_{i,t}$ and betas $\widehat{\beta}_{CBLIQ,i,t}$, $\widehat{\beta}_{EQLIQ,i,t}$, and $\widehat{\beta}_{EQ,i,t}$.
6. Calculate the Fama-MacBeth estimates as the average of the weekly regression estimates, and calculate the t -statistic of this average using Newey-West.

We repeat this entire recipe 5,000 times, which allows us to calculate the bias due to measurement error and the distribution of Fama-MacBeth estimates across simulations. The full-sample estimate is obtained simply by first averaging all expected returns, costs, and betas over all 549 weeks, and then performing a single cross-sectional regression of the average expected returns on average costs and betas.

D.2 Calibration of the simulation parameters

In the second part of this Appendix, we describe how we calibrate the parameters in the simulation setup. In general, the goal of the calibration is to match the cross-sectional and time series characteristics of the dependent and independent variables. Recall that, for each portfolio we estimate a time series of transaction costs and betas (obtained using the two-sided kernel). The key assumption that allows us to calibrate the amount of measurement error in these estimates is that the true transaction costs and betas are assumed to be constant over time in the simulation setup. Under this assumption, all time series variation in costs and betas is due to measurement error.

Table 4 reports all calibrated parameters. We first set the (risk) premium parameters λ . We choose the λ -coefficients from the benchmark specification (1) in Table 3, except for the corporate bond liquidity risk premium, which has the wrong sign according to the empirical estimates. To see if we would identify a correctly signed corporate bond liquidity risk premium, we use the coefficient in specification (6) of Table 3, where $\lambda_{CBLIQ} = -3.59$, which implies a positive contribution of the corporate bond liquidity risk premium to the expected return.

The simplest calibration step concerns the vector $\mu_{c\beta}$, which can directly be estimated by the average transaction costs and average betas across time and portfolios. Table 4 reports these. The ρ -parameters, which capture the persistence in the variables, can also be estimated easily. The autocorrelation for say, costs, is equal to $Corr(\widehat{c}_{i,t}, \widehat{c}_{i,t-k}) = \rho_c^k$. Hence we can estimate ρ_c from the autocorrelation of transaction costs. For each portfolio we focus on fitting the first 26 weekly lags, calculate $\rho_c = Corr(\widehat{c}_{i,t}, \widehat{c}_{i,t-k})^{1/k}$ for each lag, and take the average across all lags and then average across all portfolios to obtain one estimate for ρ_c . The same procedure is applied for the betas and pricing errors. As expected, Table 4 shows that there is substantial persistence at the weekly level.

To identify the measurement error variances, we compare two important statistics. The first is the time series average of the cross-sectional variance of the weekly transaction costs, in our setting given by

$$V(\widehat{c}_{i,t}) = V(c_i + u_{c,i,t}) = \Sigma_{c\beta}(1, 1) + \sigma_{u,c}^2 \quad (D12)$$

with similar expressions for the betas.

The second statistic is the cross-sectional variance of the time series average of costs (or betas), in our setting equal to

$$V(\widehat{\bar{c}}_{i,t}) = V(c_i + \overline{u_{c,i,t}}) = \Sigma_{c\beta}(1, 1) + \left(\frac{\iota' A \iota}{T^2} \right) \sigma_{u,c}^2 \quad (D13)$$

with A a $T \times T$ matrix with $A(i, j) = \rho_c^{|i-j|}$ and ι a unit vector. The factor $\frac{\iota' A \iota}{T^2}$ is smaller than 1 and captures that measurement error is (to some extent) averaged out if one calculates the time series average. Thus, the cross-sectional standard deviation of weekly costs will be higher than the cross-sectional standard deviation of the time series average of the costs, and this difference is informative about the degree of measurement error. Indeed, Table 4 shows that for all variables a positive gap between these two standard deviations exists (panel A, Std_{FB} vs. Std_{FS}). We can then calibrate the “true” variance and measurement error variance ($\Sigma_{c\beta}(1, 1)$ and $\sigma_{u,c}^2$ in case of costs, and similar for the betas) to fit these two standard deviations. The results are given in Table 4 and show that our conservative approach indeed leads to substantial measurement error variances. For the transaction costs, corporate bond liquidity betas, and equity liquidity betas, about half of the observed weekly cross-sectional variance is assigned to measurement error variance,

with the remaining half representing true cross-sectional variation. Only for the equity beta is the measurement error variance much smaller.

The next step is to identify the off-diagonal elements of $\Sigma_{c\beta}$. These are important as they capture the degree of multicollinearity. Our procedure to calibrate these is straightforward. For example, using the assumption that the measurement errors are independent across variables, for the covariance of costs and CBLIQ betas we can use the observed cross-sectional covariances as follows,

$$Cov(\widehat{c}_{i,t}, \widehat{\beta}_{CBLIQ,i,t}) = Cov(\widehat{c}_{i,t}, \widehat{\beta}_{CBLIQ,i,t}) = Cov(c_{i,t}, \beta_{CBLIQ,i,t}) = \Sigma_{c\beta}(1, 2), \quad (D14)$$

where we can thus either use the full-sample cross-sectional covariance $Cov(\widehat{c}_{i,t}, \widehat{\beta}_{CBLIQ,i,t})$, or the weekly covariances $Cov(\widehat{c}_{i,t}, \widehat{\beta}_{CBLIQ,i,t})$. We choose to use the full-sample cross-sectional covariances for two reasons. First, as discussed in the main text, the degree of multicollinearity is higher for the full-sample estimation, hence using these is conservative. Second, for the full-sample covariances most of the measurement error effects are averaged out so that it gives a more robust way to estimate the true covariance levels. Note that the setup implies that the “true” correlation between the variables is higher than the estimated correlation,²⁰

$$\begin{aligned} \left| \frac{Cov(\widehat{c}_{i,t}, \widehat{\beta}_{CBLIQ,i,t})}{\sqrt{V(\widehat{c}_{i,t})V(\widehat{\beta}_{CBLIQ,i,t})}} \right| &= \left| \frac{Cov(c_{i,t}, \beta_{CBLIQ,i,t})}{\sqrt{V(\widehat{c}_{i,t})V(\widehat{\beta}_{CBLIQ,i,t})}} \right| \\ &< \left| \frac{Cov(c_{i,t}, \beta_{CBLIQ,i,t})}{\sqrt{V(c_i)V(\beta_{CBLIQ,i,t})}} \right| = \left| \frac{\Sigma_{c\beta}(1, 2)}{\sqrt{\Sigma_{c\beta}(1, 1)\Sigma_{c\beta}(2, 2)}} \right|. \end{aligned} \quad (D15)$$

In Tables 3 and 4 we illustrate this. We report the observed correlations in panel C of Table 1 and the “true” correlations implied by the estimates for $\Sigma_{c\beta}$ in panel C of Table 4, which are higher in absolute terms. The simulation model thus builds in a substantial amount of correlation between the costs and betas.

Finally, we calibrate the variances of the “asymptotic” pricing error δ_i and the persistent pricing error $\varepsilon_{i,t}$ by matching the cross-sectional R^2 of the full-sample regression (the final specification in Table 3) and the average cross-sectional R^2 of the Fama-MacBeth regression. We cannot do this analytically and hence we perform this calibration by trial-and-error (running the simulations for several values of σ_δ^2 and σ_ε^2), which is possible since σ_δ^2 and σ_ε^2 do not enter in any of the other calibration steps. Table 4 presents the calibrated variance levels.

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²⁰ We thank an anonymous referee for pointing this out.

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